



# INTRODUCTORY COMMUTATIVE ALGEBRA



*Lecture Notes & Problem Discussions*

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## ABSTRACT

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These are lecture notes for a first course in Commutative Algebra, covering rings, ideals, and modules, together with the accompanying problem-set discussions worked out alongside the lectures. The standard references for this material are [\[AM69\]](#), [\[Bly90\]](#), and [\[DF04\]](#).

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## CHAPTER 1

# LECTURE NOTES

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### 1.1 Lecture 1

*Convention.* Rings are commutative with 1.

**Definition 1.1.** Let  $R, S$  be rings. A *morphism*  $\varphi$  from  $R$  to  $S$  is a map  $\varphi : R \rightarrow S$  such that

- (1)  $\varphi(r + s) = \varphi(r) + \varphi(s)$ ,
- (2)  $\varphi(rs) = \varphi(r)\varphi(s)$ ,
- (3)  $\varphi(1) = 1$ .

*Convention.* A subring  $R$  must contain 1.

*Question.* What is  $R[X]$ ?

$X$  is not an element of  $R$ . Consider

$$S = \{f : \mathbb{N} \rightarrow R \mid f(r) = 0 \text{ for all but finitely many } r\},$$

with the usual addition and multiplication in sequences (Cauchy product). Then  $S$  is a ring with identity

$$1_S = (1, 0, 0, \dots, 0).$$

Define  $\varphi : R \rightarrow S$  by  $\varphi(a) = (a, 0, \dots, 0)$ . Then  $\varphi$  is a ring homomorphism, for sure.

Define  $X = (0, 1, 0, \dots)$  and observe  $X^2 = (0, 0, 1, 0, \dots)$ . A polynomial  $P$  over  $R$  can then be identified as an element of  $S$ , say

$$P = (P_0, P_1, \dots, P_n, 0, \dots) = \varphi(P_0) \cdot 1_S + \varphi(P_1) \cdot X + \dots + \varphi(P_n) \cdot X^n.$$

**Definition 1.2.** Let  $R$  be a ring. A non-zero, non-unit element  $p \in R$  is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

(include the elementwise definition: for  $a, b \in R$ ,  $a \mid b$  if  $\exists r \in R$  such that  $b = r \cdot a$ ).

**Definition 1.3.** Let  $R$  be a ring. A proper ideal  $P$  of  $R$  is called a *prime ideal* if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

**Definition 1.4.** Let  $R$  be a ring.  $I \subseteq R$  is called an *ideal* if there exists a ring  $S$  and a morphism  $\varphi : R \rightarrow S$  such that  $I = \ker(\varphi)$ .

**Remark 1.5.** This immediately gives the elementwise definition of ideals.

$$\mathcal{N}(R) = \{\text{all nilpotent elements of } R\} \rightsquigarrow \text{the nil-radical of } R.$$

**Theorem 1.6.**

$$\mathcal{N}(R) = \bigcap_{P \in \text{Spec } R} P, \quad \text{where } \text{Spec } R = \text{all prime ideals of } R.$$

*Proof.* Let  $x \in \mathcal{N}(R)$ . Then for any  $P \in \text{Spec } R$ , there exists  $n_P \in \mathbb{N}$  such that

$$x^{n_P} = 0 \implies 0 \in P \implies x^{n_P} \in P \implies x \in P.$$

Now let  $x \notin \mathcal{N}(R)$ , and define

$$S = \{I \subseteq R \text{ ideal} \mid x^n \notin I \text{ for all } n\}.$$

$S \neq \emptyset$ , as  $\langle 0 \rangle \in S$ , and  $S$  is a poset (ordered by inclusion). By Zorn's Lemma, let  $M$  be a maximal element of  $S$ .

**Claim.**  $M$  is a prime ideal.

*Proof of claim.* Let  $a \notin M$ ,  $b \notin M$ . Now,

$$M + \langle a \rangle, M + \langle b \rangle \notin S \implies \exists m, n \in \mathbb{N} \text{ such that}$$

$$x^m \in M + \langle a \rangle \text{ and } x^n \in M + \langle b \rangle \implies x^{m+n} \in M + \langle ab \rangle.$$

If  $ab \in M$ , then  $x^{m+n} \in M$ , a contradiction. Hence  $ab \notin M$ , so  $M$  is prime. □

Since  $x \notin M$  and  $M \in \text{Spec } R$ , we conclude  $x \notin \bigcap_{P \in \text{Spec } R} P$ , completing the proof. □

*Question.* What are the units of  $R[X]$ ?

If  $R$  is an integral domain, then they are precisely the units of  $R$ .

**Proposition 1.7.**  $f(X) = a_n X^n + \cdots + a_1 X + a_0$  is a unit if  $a_0$  is a unit and the  $a_i$ 's are nilpotent.

*Proof.* Let  $P \in \text{Spec } R$ . We have a natural morphism  $\varphi : R[X] \rightarrow (R/P)[X]$  given by

$$\varphi(b_m X^m + \cdots + b_1 X + b_0) = \overline{b_m} X^m + \cdots + \overline{b_1} X + \overline{b_0}, \quad (\overline{b_i} = b_i \bmod P).$$

Then  $\varphi(f)$  is a unit in  $(R/P)[X]$ , and since  $R/P$  is an integral domain, this forces

$$\overline{a_i} \equiv \overline{0} \pmod{P} \text{ for all } i \in [n], \quad \text{and} \quad \overline{a_0} \text{ is a unit of } R/P.$$

That is,  $a_0 \notin P$  and  $a_1, \dots, a_n \in P$ . As  $P \in \text{Spec } R$  was arbitrary,

$$a_0 \notin P \text{ and } a_i \in P \quad \forall i \in [n], \quad \text{for all } P \in \text{Spec } R.$$

The other direction is trivial. □

## 1.2 Lecture 2

**Definition 1.8.** Let  $R$  be a ring and let  $(M, +)$  be an abelian group. A *scalar multiplication*

$$R \times M \longrightarrow M, \quad (r, m) \longmapsto rm,$$

makes  $M$  into an  $R$ -module if the following hold for all  $r, s \in R$  and  $m, m_1, m_2 \in M$ :

- (i)  $(r + s)(m) = rm + sm$ ,
- (ii)  $r(m_1 + m_2) = rm_1 + rm_2$ ,
- (iii)  $r(sm) = rsm = s(rm)$ ,
- (iv)  $1 \cdot m = m$ .

**Definition 1.9.** If  $M, N$  are  $R$ -modules, a map  $\varphi : M \rightarrow N$  is called an  *$R$ -module morphism* if

- (i)  $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$ ,
- (ii)  $\varphi(rm) = r\varphi(m)$ .

**Definition 1.10.** Let  $R$  be a ring and  $S$  a nonempty set. A *free  $R$ -module on  $S$*  is a pair  $(F, f)$ , where  $F$  is an  $R$ -module and  $f : S \rightarrow F$  is a set-theoretic map, such that: for any  $R$ -module  $M$  and any set-theoretic map  $g : S \rightarrow M$ , there exists a unique  $R$ -module morphism  $h : F \rightarrow M$

such that

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

commutes, i.e.  $h \circ f = g$ .

**Proposition 1.11.** Let  $(F, f)$  be a free  $R$ -module on  $S$ . Then:

- (1)  $f$  is injective;
- (2)  $\text{Im}(f)$  generates  $F$  as an  $R$ -module.

*Proof of (1).* Let  $x, y \in S$  with  $x \neq y$ . **Goal:**  $f(x) \neq f(y)$ .

Take any  $R$ -module  $M$  with at least two elements, and define  $g : S \rightarrow M$  such that  $g(x) \neq g(y)$ . By definition, there exists a (unique) morphism  $h : F \rightarrow M$  such that  $h \circ f = g$ . Then

$$(h \circ f)(x) = g(x) \neq g(y) = (h \circ f)(y) \implies f(x) \neq f(y). \quad \square$$

**Definition 1.12.** Let  $M$  be an  $R$ -module and  $\emptyset \neq A \subseteq M$ . The *submodule of  $M$  generated by  $A$*  is, equivalently:

- (1) the smallest submodule of  $M$  containing  $A$ ;
- (2) the intersection of all submodules of  $M$  containing  $A$ ;
- (3) the set of all finite  $R$ -linear combinations of elements of  $A$ .

*Proof of (2).* Let  $A$  be the submodule of  $F$  generated by  $\text{Im}(f)$ . Since  $f(s) \in \text{Im}(f) \subseteq A$  for every  $s \in S$ ,  $f$  factors as  $f = i_A \circ f^*$  for a set-theoretic map  $f^* : S \rightarrow A$  and the inclusion  $i_A : A \hookrightarrow F$ :

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ & \searrow f & \downarrow i_A \\ & & F \end{array}$$

As  $(F, f)$  is free on  $S$  and  $A$  is an  $R$ -module, there is a unique morphism  $h : F \rightarrow A$  such that  $h \circ f = f^*$ :

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

It is enough to show  $i_A$  is onto. We compute

$$(i_A \circ h) \circ f = i_A \circ (h \circ f) = i_A \circ f^* = f.$$

So  $i_A \circ h$  and  $\text{id}_F$  both satisfy  $(-) \circ f = f$ ; by uniqueness in the definition of freeness (applied with  $M = F, g = f$ ),  $i_A \circ h = \text{id}_F$ . Hence  $i_A$  is onto, i.e.  $A = F$ , so  $\text{Im}(f)$  generates  $F$ .  $\square$

**Theorem 1.13: Uniqueness of free modules.** Let  $(F, f)$  be a free  $R$ -module on a set  $S$ . Then  $(F', f')$  is also a free  $R$ -module on  $S$  if and only if there exists a unique  $R$ -module isomorphism  $j : F \rightarrow F'$  such that  $j \circ f = f'$ .

*Proof.* Suppose  $(F', f')$  is also free on  $S$ . Since  $(F, f)$  is free (applied to  $M = F', g = f'$ ), there is a morphism  $h : F \rightarrow F'$  with  $h \circ f = f'$ :

$$\begin{array}{ccc} S & \xrightarrow{f'} & F' \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

Likewise, since  $(F', f')$  is free (applied to  $M = F, g = f$ ), there is a morphism  $h' : F' \rightarrow F$  with  $h' \circ f' = f$ :

$$\begin{array}{ccc} S & \xrightarrow{f} & F \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Then  $(h' \circ h) \circ f = h' \circ (h \circ f) = h' \circ f' = f$ . So  $h' \circ h$  and  $\text{id}_F$  both satisfy  $(-) \circ f = f$ ; by uniqueness (with  $M = F, g = f$ ),  $h' \circ h = \text{id}_F$ . Similarly  $h \circ h' = \text{id}_{F'}$ . So  $j := h$  is an isomorphism with  $j \circ f = f'$ .

**Converse.** Suppose there is an isomorphism  $j : F \rightarrow F'$  with  $j \circ f = f'$ . To show  $(F', f')$  is free on  $S$ , take any  $R$ -module  $M$  and set-theoretic map  $g : S \rightarrow M$ ; we must produce a unique  $h' : F' \rightarrow M$  with  $h' \circ f' = g$ :

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Since  $(F, f)$  is free, there is a unique  $h : F \rightarrow M$  with  $h \circ f = g$ . From  $j \circ f = f'$  we get  $j^{-1} \circ f' = f$ , so

$$(h \circ j^{-1}) \circ f' = h \circ (j^{-1} \circ f') = h \circ f = g,$$

i.e.  $h' := h \circ j^{-1}$  works:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \nearrow f & \uparrow h \\ F' & \xrightarrow{j^{-1}} & F \end{array}$$

**Uniqueness.** Suppose  $t : F' \rightarrow M$  also satisfies  $t \circ f' = g$ . Then  $t \circ j : F \rightarrow M$  satisfies  $(t \circ$

$j) \circ f = t \circ (j \circ f) = t \circ f' = g$ , so by uniqueness of  $h$  (from freeness of  $(F, f)$ ),  $t \circ j = h$ , hence  $t = h \circ j^{-1} = h'$ .  $\square$

**Theorem 1.14: Existence of free  $R$ -modules.** For any ring  $R$  and nonempty set  $S$ , a free  $R$ -module on  $S$  exists.

*Proof.* Define

$$F = \{\theta : S \rightarrow R \mid \theta(s) = 0 \text{ for almost all } s \in S\}.$$

For  $\theta_1, \theta_2 \in F$  and  $\lambda \in R$ , define

$$(\theta_1 + \theta_2)(s) = \theta_1(s) + \theta_2(s), \quad (\lambda\theta)(s) = \lambda\theta(s).$$

One checks that  $F$  is an  $R$ -module. Define  $f : S \rightarrow F$  by  $f(s) = \delta_s$ , where  $\delta_s(t) = 1$  if  $t = s$  and 0 otherwise.

**Existence of  $h$ .** Let  $M$  be an  $R$ -module and  $g : S \rightarrow M$  a set-theoretic map. Define  $h : F \rightarrow M$  by

$$h(\theta) = \sum_{s \in S} \theta(s) g(s),$$

a finite sum for every  $\theta \in F$ . For any  $t \in S$ ,

$$(h \circ f)(t) = h(f(t)) = h(\delta_t) = \delta_t(t) g(t) = g(t),$$

so  $h \circ f = g$ :

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

**Uniqueness of  $h$ .** For  $\theta \in F$  and  $t \in S$ ,

$$\theta(t) = \theta(t) \cdot 1_R = \sum_{s \in S} \theta(s) \delta_s(t) = \left( \sum_{s \in S} \theta(s) \delta_s \right) (t) \quad \forall t \in S,$$

so  $\theta = \sum_{s \in S} \theta(s) \delta_s$ . Now let  $h' : F \rightarrow M$  be any  $R$ -module morphism with  $h' \circ f = g$ . Then for  $\theta \in F$ ,

$$h'(\theta) = h' \left( \sum_{s \in S} \theta(s) \delta_s \right) = \sum_{s \in S} \theta(s) h'(\delta_s) = \sum_{s \in S} \theta(s) h'(f(s)) = \sum_{s \in S} \theta(s) g(s) = h(\theta).$$

Hence  $h = h'$ .  $\square$

### 1.3 Lecture 3

**Theorem 1.15.** Let  $R$  be a ring such that every prime ideal is principal. Then every ideal is principal.

*Proof.* Let  $S = \{\text{all non-principal ideals of } R\}$ . For the sake of contradiction (FTSOC), suppose  $S \neq \emptyset$ . By Zorn's Lemma, let  $I$  be a maximal element of  $S$ .

Since  $I$  is not principal,  $I$  is not prime by hypothesis, so there exist  $a, b \in R$  with  $a \notin I, b \notin I$ , but  $ab \in I$ .

Let  $J = \langle a \rangle + I \supsetneq I$ ; since  $I$  is maximal among non-principal ideals,  $J \notin S$ , so  $J$  is principal, say  $J = \langle c \rangle$ .

Also let  $K = \{r \in R \mid ra \in I\} \supsetneq I$  (since  $b \in K \setminus I$ ); again  $K \notin S$ , so  $K = \langle d \rangle$  for some  $d$ .

$I \subseteq \langle cd \rangle$ . Let  $x \in I$ . Since  $I \subseteq J = \langle c \rangle$ , there is  $\lambda \in R$  with  $\lambda c = x$ . Now  $\lambda a \in I$ , so  $\lambda \in K = \langle d \rangle$ , and hence  $x \in \langle cd \rangle$ . Thus  $I \subseteq \langle cd \rangle$ .

$\langle cd \rangle \subseteq I$ . There exist  $u \in R, v \in I$  such that  $c = ua + v$  (since  $c$  generates  $J = \langle a \rangle + I$ ). Also  $da \in I$  (since  $d$  generates  $K$ ). Then

$$cd = (ua + v)d = u(ad) + vd \in I,$$

since  $ad = da \in I$  and  $v \in I$ . Hence  $\langle cd \rangle \subseteq I$ .

Together,  $I = \langle cd \rangle$ , contradicting  $I \in S$ . So  $S = \emptyset$ , i.e. every ideal of  $R$  is principal.  $\square$

**Theorem 1.16.** Let  $R$  be a ring such that every prime ideal is finitely generated. Then every ideal of  $R$  is finitely generated.

*Proof.* Let  $A = \{I \text{ ideal of } R \mid I \text{ is not finitely generated}\}$ . By Zorn's Lemma, let  $M$  be a maximal element of  $A$ .

**Claim:  $M$  is a maximal ideal.** Suppose not, so there is an ideal  $N$  with  $M \subsetneq N \subsetneq R$ . Since  $N$  is maximal (hence prime), it is finitely generated by hypothesis: there exist  $a_1, \dots, a_n \in R$  such that  $N = \langle a_1, \dots, a_n \rangle$ .

Since  $M \subsetneq N$ , some  $a_i \notin M$ . Then  $M + \langle a_i \rangle \supsetneq M$ , so by maximality of  $M$  in  $A$ ,  $M + \langle a_i \rangle \notin A$ ; hence there exist  $b_1, \dots, b_k \in R$  such that

$$M + \langle a_i \rangle = \langle b_1, \dots, b_k \rangle.$$

$\square$

**Lemma 1.17.** Let  $I, J_1, J_2$  be ideals of  $R$  such that  $I \subseteq J_1 \cup J_2$ . Then either  $I \subseteq J_1$  or  $I \subseteq J_2$ .

*Proof.* FTSOC, suppose  $I \setminus J_1 \neq \emptyset$  and  $I \setminus J_2 \neq \emptyset$ . Let  $x \in I \setminus J_1$  and  $y \in I \setminus J_2$ ; since  $I \subseteq J_1 \cup J_2$ , this forces  $x \in J_2$  and  $y \in J_1$ . Also  $x + y \in I \subseteq J_1 \cup J_2$ , so  $x + y \in J_1$  or  $x + y \in J_2$ ; WLOG  $x + y \in J_1$ . Since  $y \in J_1$ ,  $x = (x + y) - y \in J_1$ , contradicting  $x \notin J_1$ .  $\square$

**Theorem 1.18: Prime Avoidance.** Let  $I, J_1, \dots, J_n$  be ideals of  $R$  such that

- (1)  $I \subseteq J_1 \cup \dots \cup J_n$ ,
- (2)  $J_3, \dots, J_n \in \text{Spec } R$ .

Then  $I \subseteq J_i$  for some  $i$ .

*Proof.* We induct on  $n$ . The case  $n = 2$  needs no primality hypothesis and is the lemma above; assume  $n \geq 3$  and the result for any collection of  $n - 1$  such ideals.

We prove the contrapositive: if  $I \not\subseteq J_i$  for every  $i$ , then  $I \not\subseteq \bigcup_{i=1}^n J_i$ .

Fix  $i$ . Apply the induction hypothesis to  $I$  together with the  $n - 1$  ideals  $\{J_j\}_{j \neq i}$  (which still has all but at most two members prime): if  $I \subseteq \bigcup_{j \neq i} J_j$ , the induction hypothesis would give  $I \subseteq J_j$  for some  $j \neq i$ , contradicting our assumption. So  $I \not\subseteq \bigcup_{j \neq i} J_j$ , and we may choose

$$z_i \in I \setminus \bigcup_{j \neq i} J_j.$$

Since  $I \subseteq \bigcup_{i=1}^n J_i$  by hypothesis (1), and  $z_i \notin J_j$  for every  $j \neq i$ , we must have  $z_i \in J_i$ , for every  $i$ .

Set

$$z := z_1 z_2 \cdots z_{n-1} + z_n \in I$$

(a sum of an element of  $I$  and a product of elements of  $I$ , hence itself in  $I$ ). We show  $z \notin J_i$  for every  $i$ , which gives  $I \not\subseteq \bigcup_{i=1}^n J_i$ , as desired.

- If  $i \in \{1, \dots, n - 1\}$ : since  $z_i \in J_i$  is one of the factors of  $z_1 \cdots z_{n-1}$ , this product lies in  $J_i$ . If  $z \in J_i$ , then  $z_n = z - z_1 \cdots z_{n-1} \in J_i$ , contradicting  $z_n \notin J_i$  (as  $i \neq n$ ).
- If  $i = n$ : if  $z \in J_n$ , then since  $z_n \in J_n$  as well,  $z_1 \cdots z_{n-1} = z - z_n \in J_n$ . As  $J_n$  is prime, some factor  $z_k \in J_n$  for a  $k \in \{1, \dots, n - 1\}$ , contradicting  $z_k \notin J_n$  (as  $k \neq n$ ).

This completes the induction. □

**Proposition 1.19.** An element of  $\mathbb{Q}(i)$  lies in  $\mathbb{Z}[i]$  if and only if it is the root of a monic polynomial over  $\mathbb{Z}$ .

*Proof.* Let  $a + bi \in \mathbb{Q}(i)$ . Then  $a + bi$  is a root of

$$P(X) = X^2 - 2aX + (a^2 + b^2) \in \mathbb{Q}[X].$$

**Claim:**  $2a \in \mathbb{Z}$  and  $a^2 + b^2 \in \mathbb{Z}$  together imply  $a, b \in \mathbb{Z}$ .

Since  $2a \in \mathbb{Z}$ ,  $(2a)^2 \in \mathbb{Z}$ . Since  $a^2 + b^2 \in \mathbb{Z}$ ,  $4(a^2 + b^2) = (2a)^2 + (2b)^2 \in \mathbb{Z}$ , so  $(2b)^2 = 4(a^2 + b^2) - (2a)^2 \in \mathbb{Z}$ . As  $2b \in \mathbb{Q}$  has integer square,  $2b \in \mathbb{Z}$ .

Now  $(2a)^2 + (2b)^2 = 4(a^2 + b^2) \equiv 0 \pmod{4}$ . Since the square of any integer is  $\equiv 0 \pmod{4}$  if it

is even, and  $\equiv 1 \pmod{4}$  if it is odd, the only way for the sum to be  $\equiv 0 \pmod{4}$  is

$$(2a)^2 \equiv (2b)^2 \equiv 0 \pmod{4},$$

forcing  $2a, 2b$  to both be even, i.e.  $a, b \in \mathbb{Z}$ .

Since  $a + bi \in \mathbb{Z}[i]$  exactly when  $a, b \in \mathbb{Z}$ , and  $a + bi$  is a root of a monic polynomial over  $\mathbb{Z}$  exactly when  $2a, a^2 + b^2 \in \mathbb{Z}$  (the coefficients of  $P(X)$ ), the claim gives the equivalence.  $\square$

**Definition 1.20.** Let  $R \hookrightarrow S$  be rings. An element  $\alpha \in S$  is called *integral* over  $R$  if  $\alpha$  is a root of a monic polynomial

$$X^n + a_{n-1}X^{n-1} + \cdots + a_0$$

over  $R$  (i.e. with  $a_0, \dots, a_{n-1} \in R$ ).

## 1.4 Lecture 4

**Definition 1.21.** Let  $R$  be a subring of  $S$ , and let  $U \subseteq S$  be a subset. Define

$$R[U] := \text{the smallest subring of } S \text{ containing } R \text{ and } U.$$

**Remark 1.22.** For  $U, V \subseteq S$ ,

$$R[U][V] = R[U \cup V] = R[V][U].$$

In particular, if  $V = \{u_1, \dots, u_n\}$ , then

$$R[u_1, \dots, u_n] = R[u_1][u_2] \cdots [u_n].$$

**Proposition 1.23.**

$$R[u] = \{a_0 + a_1u + \cdots + a_nu^n \mid a_i \in R, n \in \mathbb{N}_{\geq 0}\}.$$

*Proof.* Let  $T = \{a_0 + a_1u + \cdots + a_nu^n \mid a_i \in R, n \in \mathbb{N}_{\geq 0}\}$ .

$T$  is a **subring** of  $S$ . It contains 1 (take  $n = 0, a_0 = 1$ ). It is closed under addition: given two such expressions, pad the shorter one with zero coefficients to a common degree and add coefficientwise, which stays in  $R$  since  $R$  is a ring. It is closed under multiplication: expanding a product of two such expressions and collecting powers of  $u$ , each resulting coefficient is a finite sum of products of elements of  $R$ , hence lies in  $R$ .

$R[u] \subseteq T$ . Since  $R \subseteq T$  (take  $n = 0$ ) and  $u \in T$  (take  $n = 1, a_0 = 0, a_1 = 1$ ), and  $T$  is a subring of  $S$  containing  $R$  and  $u$ , while  $R[u]$  is by definition the *smallest* such subring, we get  $R[u] \subseteq T$ .

$T \subseteq R[u]$ . Since  $R[u]$  is a subring containing  $R$  and  $u$ , it is closed under addition and multiplication. So for any  $a_0, \dots, a_n \in R \subseteq R[u]$ , each  $a_iu^i \in R[u]$ , and hence  $a_0 + a_1u + \cdots + a_nu^n \in R[u]$ .

Combining both containments,  $R[u] = T$ . □

**Example 1.24.** The map  $\mathbb{Z}[X] \rightarrow \mathbb{Z}[i]$ ,  $X \mapsto i$ , is a surjective ring morphism with kernel  $\langle X^2 + 1 \rangle$ .

**Proposition 1.25.** Let  $R, S$  be rings and  $\varphi : R \rightarrow S$  be a morphism. Let  $\alpha \in S$  be an element. Then  $\varphi$  can be uniquely extended to a morphism  $\tilde{\varphi} : R[X] \rightarrow S$  such that  $\tilde{\varphi}|_R = \varphi$  and  $\tilde{\varphi}(X) = \alpha$ .

*Proof sketch.* Define  $\tilde{\varphi} : R[X] \rightarrow S$  by

$$\tilde{\varphi}(a_0 + a_1X + \cdots + a_nX^n) = \varphi(a_0) + \varphi(a_1)\alpha + \cdots + \varphi(a_n)\alpha^n,$$

and  $\tilde{\varphi}(A \cdot B) = \tilde{\varphi}(A) \cdot \tilde{\varphi}(B)$ . □

Let  $R$  be a subring of  $S$ , and let  $u \in S$ . Then there is a unique morphism  $\varphi : R[X] \rightarrow S$  such that  $\varphi|_R = \text{id}_R$  and  $\varphi(X) = u$  (apply the above with  $\varphi = \text{id}_R$ ,  $\alpha = u$ ). Then  $\text{Im}(\varphi) = R[u]$ ; letting  $I = \ker \varphi$ ,

$$R[X]/I \cong R[u].$$

*Notation.*  $I \cap R = \langle 0 \rangle$ .

Conversely, let  $I$  be an ideal of  $R[X]$  such that  $I \cap R = \langle 0 \rangle$ . We have the natural projection map

$$R[X] \longrightarrow R[X]/I.$$

Since  $I \cap R = 0$ ,  $R \hookrightarrow R[X]/I$ , so we can identify  $a \in R$  with its image  $a + I$ .

*Question.* What is the image of  $R[X]$  in  $R[X]/I$ ? (It will be determined by  $\text{Im}(R)$  and that of  $X$ .)

Under the projection,  $X \mapsto X + I$ . Writing  $u = X + I$ ,

$$R[X]/I \cong R[u].$$

**Definition 1.26.** Let  $R \hookrightarrow S$ .  $S$  is said to be an *integral extension* of  $R$  if each  $\alpha \in S$  is integral over  $R$ .

*Notation.* (Recall.) Let  $R$  be a commutative ring with 1, and let  $A \in M_n(R)$ . Then

$$A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \det(A) \cdot I_n.$$

**Theorem 1.27.** Let  $R \hookrightarrow S$  and  $b \in S$ . Then

$$b \text{ is integral over } R \iff R[b] \text{ is a finitely generated } R\text{-module}.$$

*Proof.* ( $\Rightarrow$ ) Assume  $b$  is integral over  $R$ ; then there is a monic polynomial  $f(X) \in R[X]$  such that

$f(b) = 0$ . Let  $\deg f = n \geq 1$ . Take any  $g(b) \in R[b]$ . By the division algorithm,

$$g(X) = q(X)f(X) + r(X), \quad \deg r < n,$$

so  $g(b) = r(b)$ . Hence  $R[b]$  is generated (as an  $R$ -module) by  $\{1, b, \dots, b^{n-1}\}$ .

( $\Leftarrow$ ) Conversely, let  $R[b]$  be a finitely generated  $R$ -module, say  $R[b] = \langle w_1, \dots, w_n \rangle$ . Then for each  $i \in [n]$ ,

$$bw_i = a_{i1}w_1 + \dots + a_{in}w_n, \quad a_{ij} \in R,$$

i.e.  $(bI - A)\vec{w} = 0$ , where  $A = (a_{ij})$ . Multiplying by  $\text{adj}(bI - A)$ ,

$$\det(bI - A)w_i = 0 \quad \forall i \in [n].$$

Since  $1 \in R[b] = \langle w_1, \dots, w_n \rangle$ , we get  $\det(bI - A) = 0$ , which is a monic equation in  $b$  (of degree  $n$ ). Hence  $b$  is integral over  $R$ .  $\square$

**Theorem 1.28.** Let  $R \hookrightarrow S$ , and let  $b_1, b_2, \dots, b_n \in S$ . Then each  $b_i$  is integral over  $R$  if and only if  $R[b_1, \dots, b_n]$  is a finitely generated  $R$ -module.

*Proof.* Induct on  $n$ . The case  $n = 1$  is the previous theorem.

For  $n > 1$ : assume  $b_1, \dots, b_n$  are integral over  $R$ . In particular  $b_n$  is integral over  $R$ , hence also integral over  $R[b_1, \dots, b_{n-1}]$ ; by the previous theorem (applied over the base ring  $R[b_1, \dots, b_{n-1}]$ ),  $R[b_1, \dots, b_{n-1}][b_n]$  is a finitely generated  $R[b_1, \dots, b_{n-1}]$ -module, which (combined with the induction hypothesis that  $R[b_1, \dots, b_{n-1}]$  is a finitely generated  $R$ -module) makes  $R[b_1, \dots, b_n]$  a finitely generated  $R$ -module.

The converse is the same as the  $n = 1$  case.  $\square$

**Exercise 1.29.** Let  $R \hookrightarrow S$ , and let  $a, b \in S$  be integral over  $R$ . Then  $a \pm b$  and  $ab$  are integral over  $R$ .

## 1.5 Lecture 5

**Corollary 1.30.** Let  $R \hookrightarrow S$  and  $b_1, \dots, b_n \in S$  be integral over  $R$  (equivalently,  $R[b_1, \dots, b_n]$  is a finitely generated  $R$ -module). Then every  $b \in R[b_1, \dots, b_n]$  is integral over  $R$ .

*Proof.* Since  $b \in R[b_1, \dots, b_n]$ ,

$$R[b_1, \dots, b_n] = R[b_1, \dots, b_n, b],$$

which is a finitely generated  $R$ -module. By the previous theorem,  $b$  is integral over  $R$ .  $\square$

**Proposition 1.31.** Let  $A \subseteq B \subseteq C$  be rings such that  $B$  is integral over  $A$  and  $C$  is integral over  $B$ . Then  $C$  is integral over  $A$ .

*Proof.* Let  $c \in C$ . Then  $c$  is integral over  $B$ :

$$c^n + b_{n-1}c^{n-1} + \cdots + b_0 = 0, \quad b_i \in B.$$

Let  $R := A[b_0, \dots, b_{n-1}]$ . Then  $c$  is integral over  $R$ , so  $R[c]$  is a finitely generated  $R$ -module.

Since  $b_0, \dots, b_{n-1}$  are integral over  $A$ ,  $R$  is a finitely generated  $A$ -module. Hence  $R[c]$  is a finitely generated  $A$ -module, and since  $c \in R[c]$ ,  $c$  is integral over  $A$ .  $\square$

**Definition 1.32.** Define  $\bar{R}^S := \{r \in S \mid r \text{ is integral over } R\}$ . Note that  $\bar{R}^S$  is a subring of  $S$ . This  $\bar{R}^S$  is called the *integral closure* of  $R$  in  $S$ .

**Definition 1.33.**  $R$  is said to be *integrally closed* in  $S$  if  $\bar{R}^S = R$ .

$$R \hookrightarrow \bar{R}^S \hookrightarrow S.$$

**Observation 1.34.**  $\bar{R}^S$  is integrally closed in  $S$ .

**Definition 1.35.** Let  $R$  be an integral domain with quotient field  $F = \mathbb{Q}(R)$ . We say that  $R$  is *integrally closed* if  $R$  is integrally closed in  $F$ .

**Proposition 1.36.** If  $R$  is a UFD, then  $R$  is integrally closed.

*Proof.* Let  $F = \mathbb{Q}(R)$ . Say  $r/s \in F$  with  $r, s \in R$ ,  $s \neq 0$ , is integral over  $R$ , and suppose  $r, s$  have no common prime factors. Then

$$\left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots + a_1\left(\frac{r}{s}\right) + a_0 = 0, \quad a_i \in R,$$

so

$$r^n + a_{n-1}r^{n-1}s + \cdots + a_1rs^{n-1} + a_0s^n = 0.$$

Since  $s$  divides every term but the first,  $s \mid r^n$ . FTSOC, if  $s$  is not a unit, it has a prime factor  $p$ ; then  $p \mid r^n$ , and since  $p$  is prime,  $p \mid r$  – contradicting that  $r, s$  have no common prime factors. Hence  $s$  is a unit in  $R$ , so  $r/s \in R$ .  $\square$

**Example 1.37.**  $\mathbb{Z}[\sqrt{-5}]$  is integrally closed but not a UFD. (Exercise.)

**Example 1.38.**  $\mathbb{Z}[\sqrt{5}]$  is an integral domain which is not integrally closed.

*Proof.*  $\mathbb{Q}(\mathbb{Z}[\sqrt{5}]) = \mathbb{Q}(\sqrt{5})$ , and

$$\frac{1 + \sqrt{5}}{2} \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Z}[\sqrt{5}]$$

is a root of the monic polynomial  $X^2 - X - 1 \in \mathbb{Z}[X]$ , hence integral over  $\mathbb{Z}[\sqrt{5}]$ , yet not itself an element of  $\mathbb{Z}[\sqrt{5}]$ .  $\square$

## Extensions and Contractions

Let  $\varphi : R \rightarrow S$  be a ring morphism. Then  $\varphi$  factors as

$$R \xrightarrow{\varphi} \varphi(R) \xhookrightarrow{i} S \quad \text{with} \quad R / \text{Ker } \varphi \cong \varphi(R).$$

- (1) If  $I \subseteq R$  is an ideal,  $\varphi(I)$  need *not* be an ideal of  $S$ .
- (2) If  $J \subseteq S$  is an ideal,  $\varphi^{-1}(J)$  is an ideal of  $R$ .
- (3) If  $Q \in \text{Spec } S$ , then  $\varphi^{-1}(Q) \in \text{Spec } R$ .

*Question.* Suppose  $P \in \text{Spec } R$ . Can we say that  $\langle \varphi(P) \rangle$  is prime in  $S$ ?

Consider  $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]: \langle 2 \rangle \mapsto \langle 1+i \rangle \langle 1-i \rangle$ . If  $p \equiv 1 \pmod{4}$  is prime, then  $\langle p \rangle$  is *not* prime in  $\mathbb{Z}[i]$  (so the answer to the question above is no, in general).

**Exercise 1.39.** If  $p \equiv 3 \pmod{4}$ , then  $\langle p \rangle$  is a prime ideal in  $\mathbb{Z}[i]$ .

**Definition 1.40.** Let  $\varphi : R \rightarrow S$  be a ring morphism.

- For  $I$  an ideal of  $R$ , the *extension*  $I^e$  is the ideal of  $S$  generated by  $\varphi(I)$ .
- For  $J$  an ideal of  $S$ , the *contraction*  $J^c := \varphi^{-1}(J)$ , an ideal of  $R$ .

**Exercise 1.41.**

$$I \subseteq I^{ec}, \quad J \supseteq J^{ce}, \quad I^e = I^{ece}, \quad J^c = J^{cec}.$$

## Rings of Fractions

**Definition 1.42.** Let  $R$  be a ring. A subset  $S \subseteq R$  is called *multiplicative* if

- (1)  $1 \in S$ ,
- (2)  $a, b \in S \implies ab \in S$ .

**Construction 1.43.** On  $R \times S$ , define the relation  $\sim$  by

$$(a, s) \sim (b, t) \iff \exists u \in S \text{ such that } u(at - bs) = 0.$$

$\sim$  is transitive. Suppose  $(a, s) \sim (b, t)$  and  $(b, t) \sim (c, u)$ ; so  $(at - bs)v = 0$  for some  $v \in S$ , and  $(bu - ct)w = 0$  for some  $w \in S$ . Then

$$atv - bsv = 0 \rightsquigarrow uw \cdot atv - bsuw = 0, \quad buw - ctw = 0 \rightsquigarrow buwsv - ctwsv = 0.$$

Adding,

$$(au - cs)vtw = 0, \quad vtw \in S,$$

so  $(a, s) \sim (c, u)$ . □

*Notation.* The equivalence class of  $(a, s)$  is denoted  $\frac{a}{s}$ .

Let

$$S^{-1}R := \{\text{all equivalence classes}\} = \left\{ \frac{a}{s} \mid a \in R, s \in S \setminus \{0\} \right\}.$$

Define

$$\frac{a}{s} + \frac{b}{t} := \frac{at + bs}{st} \quad (\text{well-defined}), \quad \frac{a}{s} \cdot \frac{b}{t} := \frac{ab}{st} \quad (\text{well-defined}).$$

$S^{-1}R$  is a commutative ring with 1. We have a canonical ring morphism

$$\varphi : R \rightarrow S^{-1}R, \quad r \mapsto \frac{r}{1}.$$

**Remark 1.44.**  $\varphi$  is not injective in general.

**Proposition 1.45.** Let  $g : R \rightarrow A$  be a ring morphism such that  $g(S) \subseteq A^\times$  (the units of  $A$ ). Then there is a unique morphism  $h : S^{-1}R \rightarrow A$  such that

$$\begin{array}{ccc} R & \xrightarrow{g} & A \\ \varphi \downarrow & \nearrow h & \\ S^{-1}R & & \end{array}$$

commutes, i.e.  $h \circ \varphi = g$ .

*Proof. Uniqueness.* Suppose such an  $h$  exists. Then

$$h\left(\frac{r}{1}\right) = h(\varphi(r)) = g(r), \quad h\left(\frac{1}{s}\right) = h\left(\left(\frac{s}{1}\right)^{-1}\right) = h\left(\frac{s}{1}\right)^{-1} = g(s)^{-1},$$

so

$$h\left(\frac{r}{s}\right) = h\left(\frac{r}{1}\right)h\left(\frac{1}{s}\right) = g(r)g(s)^{-1},$$

i.e.  $h$  is uniquely determined by  $g$ .

**Existence.** Define  $h : S^{-1}R \rightarrow A$  by  $h(r/s) = g(r)g(s)^{-1}$ .

*Well-definedness.* Suppose  $r/s = r'/s'$ ; then there is  $t \in S$  with  $t(rs' - r's) = 0$ , so

$$g(t)(g(r)g(s') - g(r')g(s)) = 0.$$

Since  $g(t) \in A^\times$ ,  $g(r)g(s') = g(r')g(s)$ , i.e.  $g(r)g(s)^{-1} = g(r')g(s')^{-1}$ . □

**Example 1.46.** Let  $P \in \text{Spec } R$  and  $S = R \setminus P$ . Then  $S$  is multiplicative; write  $S^{-1}R =: R_P$ . Let

$$PR_P := \left\{ \frac{p}{s} \mid p \in P, s \in R \setminus P \right\},$$

an ideal of  $R_P$ . If  $b/t \notin PR_P$ , then  $b \notin P$ , so  $b \in R \setminus P$ , so  $b/t$  is a unit in  $R_P$ . Hence  $PR_P$  is the unique maximal ideal of  $R_P$ , so  $R_P$  is local.

**Example 1.47.** Let  $f \in R$  and  $S = \{1, f, f^2, \dots\}$ . Then  $R_f := S^{-1}R$ .

**Construction 1.48.** Let  $R$  be a ring,  $S$  a multiplicative subset of  $R$ , and  $M$  an  $R$ -module. On  $M \times S$ , define  $\sim$  by

$$(m, s) \sim (m', s') \iff \exists t \in S \text{ such that } t(ms' - m's) = 0.$$

Then  $S^{-1}M := (M \times S)/\sim$  is an  $S^{-1}R$ -module.

Let  $M, N$  be  $R$ -modules and  $\varphi : M \rightarrow N$  an  $R$ -module morphism. Define  $S^{-1}\varphi : S^{-1}M \rightarrow S^{-1}N$  by

$$\frac{m}{s} \mapsto \frac{\varphi(m)}{s}.$$

$S^{-1}\varphi$  is a well-defined  $S^{-1}R$ -module morphism.

**Proposition 1.49.**  $S^{-1}(-)$  preserves exactness: if

$$M' \xrightarrow{\varphi} M \xrightarrow{\psi} M''$$

is exact at  $M$ , then

$$S^{-1}M' \xrightarrow{S^{-1}\varphi} S^{-1}M \xrightarrow{S^{-1}\psi} S^{-1}M''$$

is exact at  $S^{-1}M$ .

*Proof.* We have  $\psi \circ \varphi = 0$ , so  $(S^{-1}\psi) \circ (S^{-1}\varphi) = S^{-1}(\psi \circ \varphi) = 0$ . Hence  $\text{Im}(S^{-1}\varphi) \subseteq \text{Ker}(S^{-1}\psi)$ .

Conversely, let  $m/s \in \text{Ker}(S^{-1}\psi)$ . Then  $\psi(m)/s = 0$ , so there is  $t \in S$  with  $t\psi(m) = 0$ , i.e.

$\psi(tm) = 0$ , i.e.  $tm \in \text{Ker}(\psi) = \text{Im}(\varphi)$ . So there is  $m' \in M'$  with  $\varphi(m') = tm$ . Then

$$\frac{m}{s} = \frac{mt}{st} = \frac{\varphi(m')}{st} = (S^{-1}\varphi)\left(\frac{m'}{st}\right) \in \text{Im}(S^{-1}\varphi).$$

Hence  $\text{Ker}(S^{-1}\psi) \subseteq \text{Im}(S^{-1}\varphi)$ , giving equality.  $\square$

## 1.6 Lecture 6

**Theorem 1.50.** Let  $M$  be an  $R$ -module generated by  $n$  elements, and let

$$\varphi \in \text{Hom}(M, M) := \{f : M \rightarrow M \mid f \text{ is an } R\text{-module morphism}\}.$$

Let  $I$  be an ideal of  $R$  such that  $\varphi(M) \subseteq IM$ . Then there is a relation of the form

$$\varphi^n + a_{n-1}\varphi^{n-1} + \cdots + a_0 = 0, \quad a_i \in I^i.$$

*Proof.* Let  $M = Rw_1 + \cdots + Rw_n$ , so  $IM = Iw_1 + \cdots + Iw_n$ ; in particular  $\varphi(M) \subseteq IM$ . Write

$$\varphi(w_i) = \sum_{j=1}^n a_{ij}w_j, \quad a_{ij} \in I \text{ for all } i,$$

i.e.

$$\sum_{j=1}^n (\varphi \delta_{ij} - a_{ij})w_j = 0.$$

The coefficient matrix has  $(i, j)$ -entry  $\varphi \delta_{ij} - a_{ij}$ , with entries in the commutative subring of  $\text{End}(M)$  generated by  $R$  (acting by scalars) and  $\varphi$ . Let  $d = \det(\text{coefficient matrix})$ . Multiplying by the adjugate matrix,

$$d \cdot w_k = 0 \quad \forall k \in [n],$$

so  $d \cdot M = 0$ , i.e.  $d$  is the zero endomorphism of  $M$ . Expanding the determinant gives exactly a relation of the stated form.  $\square$

**Theorem 1.51: Nakayama's Lemma.** Let  $M$  be a finitely generated  $R$ -module and  $I$  an ideal of  $R$ . If  $M = IM$ , then there exists  $a \in R$  such that  $aM = 0$  and  $a \equiv 1 \pmod{I}$ . Furthermore, if  $I$  is the Jacobson radical of  $R$ , then  $M = 0$ .

*Proof.* Take  $\varphi = \text{id}_M$ . By the previous theorem, we have

$$1 + a_{n-1} + \cdots + a_0 = 0 \quad (\text{as endomorphisms of } M).$$

Define  $a := 1 + a_{n-1} + \cdots + a_0$ . Then  $aM = 0$  and  $a \equiv 1 \pmod{I}$ .

If  $I \subseteq J(R)$  (the Jacobson radical), then  $a - 1 \in J(R)$ , so  $a$  is a unit; hence  $aM = 0$  forces  $M = 0$ .

$$a^{-1}(aM) = 0. \quad \square$$

**Theorem 1.52.** Let  $M$  be a finitely generated  $R$ -module and  $I$  an ideal of  $R$  such that  $M = IM$ . Then there exists  $i \in I$  such that  $(1 + i)M = 0$ .

*Proof (Ojanguren).* Induct on the minimal number of generators  $\nu(M) = n$  of  $M$ .

If  $\nu(M) = 0$ , then  $M = 0$  and we are done.

Let  $M = Rm_1 + \cdots + Rm_n$ , and set  $M' := M/Rm_n$ . Then

$$IM' = \frac{IM + Rm_n}{Rm_n} = \frac{M + Rm_n}{Rm_n} = \frac{M}{Rm_n} = M',$$

using  $IM = M$ . Since  $\nu(M') < n$ , by induction there is  $x \in I$  such that  $(1 + x)M' = 0$ , i.e.

$$(1 + x) \frac{M}{Rm_n} = 0 \implies (1 + x)M \subseteq Rm_n.$$

Then

$$(1 + x)M = (1 + x)IM = I(1 + x)M \subseteq I \cdot Rm_n = Im_n,$$

using  $(1 + x)M \subseteq Rm_n$  in the last containment. In particular  $(1 + x)m_n \in Im_n$ , so  $(1 + x)m_n = ym_n$  for some  $y \in I$ , giving

$$(1 + x - y)m_n = 0.$$

Now for any  $\mu \in M$ ,  $(1 + x)\mu = rm_n$  for some  $r \in R$  (since  $(1 + x)M \subseteq Rm_n$ ), so

$$(1 + x - y)(1 + x)\mu = r(1 + x - y)m_n = 0.$$

Hence  $(1 + x)(1 + x - y)M = 0$ . Expanding,  $(1 + x)(1 + x - y) = 1 + i$  where  $i := x + (x - y) + x(x - y) \in I$ . So  $(1 + i)M = 0$ .  $\square$

**Definition 1.53.** Let  $M, N$  be  $R$ -modules. The *tensor product* of  $M, N$  is a pair  $(T, \theta)$  such that  $T$  is an  $R$ -module,  $\theta : M \times N \rightarrow T$  is bilinear, and for every bilinear map  $f : M \times N \rightarrow P$ , there is a unique linear map  $h : T \rightarrow P$  such that

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \theta \downarrow & \nearrow h & \\ T & & \end{array}$$

commutes, i.e.  $h \circ \theta = f$ .

Recall the canonical morphism  $R \rightarrow S^{-1}R$ ,  $r \mapsto r/1$ , and let  $I$  be an ideal of  $R$ . An element of the

extended ideal  $I^e$  is a finite sum

$$\frac{r_1}{s_1} \cdot \frac{x_1}{1} + \cdots + \frac{r_n}{s_n} \cdot \frac{x_n}{1} = \frac{s'_1 r_1 x_1 + \cdots + s'_n r_n x_n}{s_1 \cdots s_n},$$

where  $r_i \in R$ ,  $s_i \in S$ ,  $x_i \in I$  (writing  $s'_i$  for the product of all  $s_j$  with  $j \neq i$ ).

**Proposition 1.54.**

- (1) Every ideal of  $S^{-1}R$  is an extended ideal.
- (2) The prime ideals of  $S^{-1}R$  are in one-to-one correspondence,  $P \leftrightarrow S^{-1}P$ , with the prime ideals  $P$  of  $R$  such that  $S \cap P = \emptyset$ .

*Proof.* See [AM69]. □

### Local Properties of Modules

**Proposition 1.55.** Let  $M$  be an  $R$ -module. Then the following are equivalent:

- (1)  $M = 0$ ;
- (2)  $M_P = 0$  for all  $P \in \text{Spec}(R)$ ;
- (3)  $M_{\mathfrak{m}} = 0$  for all  $\mathfrak{m} \in \text{Max}(R)$ .

(Here for  $P \in \text{Spec}(R)$ , writing  $S = R \setminus P$ , we set  $M_P := S^{-1}M$ .)

*Proof.* (1)  $\Rightarrow$  (2)  $\Rightarrow$  (3) trivially.

(3)  $\Rightarrow$  (1): Let  $M \neq 0$ ; then there is  $x \in M$  with  $x \neq 0$ . Let  $I = \text{Ann}(x) = \{r \in R \mid rx = 0\}$ , a proper ideal of  $R$ . Let  $\mathfrak{m}$  be a maximal ideal with  $I \subseteq \mathfrak{m}$ .

Consider  $x/1 \in M_{\mathfrak{m}}$ . Since  $M_{\mathfrak{m}} = 0$ ,  $x/1 = 0$ , so there is  $t \in R \setminus \mathfrak{m}$  with  $tx = 0$ . Then  $t \in I \subseteq \mathfrak{m}$  – contradicting  $t \notin \mathfrak{m}$ . □

**Exercise 1.56.** Let  $\varphi \in \text{Hom}(M, N)$ . TFAE:

- (1)  $\varphi$  is injective (resp. surjective);
- (2)  $\varphi_P : M_P \rightarrow N_P$  is injective (resp. surjective) for all  $P \in \text{Spec}(R)$ ;
- (3)  $\varphi_{\mathfrak{m}} : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$  is injective (resp. surjective) for all  $\mathfrak{m} \in \text{Max}(R)$ .

**Proposition 1.57.** Let  $R \hookrightarrow S$  be rings with  $S$  integral over  $R$ .

- (1) If  $J$  is an ideal of  $S$ , then  $R/(J \cap R) \hookrightarrow S/J$  is integral.
- (2) If  $T$  is a multiplicative subset of  $R$ , then  $T^{-1}R \hookrightarrow T^{-1}S$  is integral.

*Proof.* (1) is straightforward (reduce the monic integral equation modulo  $J$ ).

(2) Let  $x/t \in T^{-1}S$  ( $x \in S, t \in T$ ). Since  $x$  is integral over  $R$ ,

$$x^n + a_1x^{n-1} + \cdots + a_n = 0 \quad \text{for some } a_i \in R.$$

Dividing by  $t^n$ ,

$$\left(\frac{x}{t}\right)^n + \frac{a_1}{t} \left(\frac{x}{t}\right)^{n-1} + \cdots + \frac{a_n}{t^n} = 0,$$

so  $x/t$  is integral over  $T^{-1}R$ . □

**Proposition 1.58.** Let  $R \hookrightarrow S$  be an integral extension, with  $R, S$  both integral domains. Then  $R$  is a field if and only if  $S$  is a field.

*Proof.* ( $\Rightarrow$ ) Let  $R$  be a field and  $y \in S \setminus \{0\}$ . Since  $y$  is integral over  $R$ , take a monic relation of minimal degree,

$$y^n + a_1y^{n-1} + \cdots + a_n = 0, \quad a_i \in R.$$

If  $a_n = 0$ , then  $y(y^{n-1} + a_1y^{n-2} + \cdots + a_{n-1}) = 0$ ; since  $S$  is a domain and  $y \neq 0$ , this gives a relation of degree  $n - 1$ , contradicting minimality. So  $a_n \neq 0$ , and as  $R$  is a field,  $a_n$  is a unit. Then

$$y(y^{n-1} + \cdots + a_{n-1}) = -a_n,$$

and since  $-a_n$  is a unit,  $y$  is a unit.

( $\Leftarrow$ ) Let  $S$  be a field. Take any  $x \in R \setminus \{0\}$ . Then  $x^{-1} \in S$  is integral over  $R$ :

$$x^{-m} + a_1x^{-(m-1)} + \cdots + a_m = 0, \quad a_i \in R,$$

so

$$x^{-1} = -(a_1 + a_2x + \cdots + a_mx^{m-1}) \in R.$$

□

## 1.7 Lecture 7

**Example 1.59.** Let  $k$  be a field,  $t$  an indeterminate, and  $A = k[t^2, t^3] \subseteq B = k[t]$ . Since  $A$  is a subring of the integral domain  $B$ ,  $A$  is an integral domain. The fraction field of both  $A$  and  $B$  is  $k(t)$ , and since  $B$  is a UFD,  $B$  is integrally closed in  $k(t)$ .

*Question.* Is  $A$  integrally closed in  $k(t)$ ?

No:  $t$  is integral over  $A$ , being a root of  $X^2 - t^2 \in A[X]$ , but  $t \notin A$ . So  $A$  is not integrally closed.

*Question.* What is the integral closure of  $A$  in  $k(t)$ ?

It is  $B$ , since  $B$  is integrally closed in  $k(t)$ .

**Example 1.60.** Similarly,

$$\begin{array}{ccc} B & \hookrightarrow & \mathbb{Q}(i) \\ \uparrow & & \uparrow \\ \mathbb{Z} & \hookrightarrow & \mathbb{Q} \end{array}$$

where  $B$  is the integral closure of  $\mathbb{Z}$  (an integrally closed domain) in  $\mathbb{Q}(i)$ . One checks  $B = \mathbb{Z}[i]$ :  $\min(i, \mathbb{Q}) = X^2 + 1 \in \mathbb{Z}[X]$ .

**Theorem 1.61.** Let  $A$  be an integrally closed domain with  $K = \mathbb{Q}(A)$ , and let  $L$  be an algebraic extension of  $K$ . Then  $\alpha \in L$  is integral over  $A$  if and only if  $\min(\alpha, K) \in A[X]$ .

*Proof.* ( $\Leftarrow$ ) is trivial: if  $f(X) = X^n + a_1X^{n-1} + \cdots + a_n = \min(\alpha, K) \in A[X]$ , then  $f(\alpha) = 0$  exhibits  $\alpha$  as integral over  $A$ .

( $\Rightarrow$ ) Let  $\alpha$  be integral over  $A$ ; we show the coefficients  $a_i$  of  $f := \min(\alpha, K)$  lie in  $A$ .

Let  $\bar{L}$  be an algebraic closure of  $L$ . In  $\bar{L}$ ,  $f$  splits as

$$f(X) = (X - \alpha_1) \cdots (X - \alpha_n),$$

where the  $\alpha_i$  are the conjugates of  $\alpha$  over  $K$ . For each  $i$  there is a  $K$ -isomorphism  $K[\alpha] \cong K[\alpha_i]$  fixing  $K$  and sending  $\alpha \mapsto \alpha_i$ .

Since  $\alpha$  is integral over  $A$ , there is a monic  $g(X) \in A[X]$  with  $g(\alpha) = 0$ ; as  $f = \min(\alpha, K)$ ,  $f \mid g$  in  $K[X]$ . Applying the isomorphism above,  $g(\alpha_i) = 0$  for each  $i$ , so each  $\alpha_i$  is integral over  $A$ .

Now  $a_1, \dots, a_n \in A[\alpha_1, \dots, \alpha_n]$  (they are, up to sign, the elementary symmetric functions of the  $\alpha_i$  – Newton's identities), and each  $\alpha_i$  is integral over  $A$ , so each  $a_i$  is integral over  $A$ . But  $a_i \in K$  and  $A$  is integrally closed in  $K$ , so  $a_i \in A$  for all  $i$ .  $\square$

**Exercise 1.62.**

- (1)  $k[X, Y] / \langle Y^2 - X^3 \rangle \cong k[t^2, t^3]$  (the coordinate ring of the curve  $Y^2 = X^3$ ).
- (2) Show directly that  $k[t^2, t^3]$  is not a UFD.

**Example 1.63.** Let  $A$  be a UFD with  $2 \in A^\times$ , and let  $f \in A$  be square-free. Then  $A[\sqrt{f}]$  is an integrally closed domain.

*Proof.* Since  $\langle X^2 - f \rangle \in \text{Spec}(A[X])$  (as  $f$  is square-free,  $X^2 - f$  is irreducible),

$$A[\sqrt{f}] \cong A[X] / \langle X^2 - f \rangle$$

is an integral domain.

Let  $\alpha$  be a square root of  $f$ ; we ask whether  $A[\alpha]$  is integrally closed. Let  $K := \mathbb{Q}(A)$ ; since  $A$  is a

UFD,  $A$  is integrally closed in  $K$ .

**Case  $\alpha \in K$ :** Then  $X^2 - f \in A[X]$  has root  $\alpha \in K$ , so  $\alpha$  is integral over  $A$ , hence  $\alpha \in A$  (as  $A$  is integrally closed in  $K$ ). So  $A[\alpha] = A$ , trivially integrally closed.

**Case  $\alpha \notin K$ :** Then

$$\mathbb{Q}(A[\alpha]) = K(\alpha) = K + K\alpha,$$

and every  $\theta \in K(\alpha)$  can be written uniquely as  $\theta = x + y\alpha$  with  $x, y \in K$ ; its minimal polynomial over  $K$  is

$$X^2 - 2xX + (x^2 - fy^2).$$

Suppose  $\theta$  is integral over  $A[\alpha]$ . Since  $\alpha$  is integral over  $A$  (a root of  $X^2 - f \in A[X]$ ),  $A[\alpha]$  is integral over  $A$ ; by transitivity,  $\theta$  is integral over  $A$ . By the previous theorem (as  $A$  is integrally closed in  $K$ ),

$$X^2 - 2xX + (x^2 - fy^2) \in A[X],$$

i.e.  $2x \in A$  and  $x^2 - fy^2 \in A$ . Since 2 is a unit,  $x \in A$ ; hence  $fy^2 = x^2 - (x^2 - fy^2) \in A$ .

If some prime  $p \in A$  divides the denominator of  $y$  (writing  $y \in K$  in lowest terms), then since  $fy^2 \in A$  and  $f$  is square-free, we would need  $p^2 \mid f$  – a contradiction. Hence  $y \in A$ , so  $\theta = x + y\alpha \in A[\alpha]$ .

Together with the trivial case  $A[\alpha] = A$ , this shows  $A[\alpha]$  is integrally closed.  $\square$

**Lemma 1.64.** Let  $A \hookrightarrow B$  be an integral extension. If  $P \in \text{Max}(B)$ , then  $P \cap A \in \text{Max}(A)$ . Conversely, if  $\mathfrak{p} \in \text{Max}(A)$ , then there exists  $P \in \text{Spec}(B)$  such that  $P \cap A = \mathfrak{p}$  (and  $P$  is maximal too).

*Proof.*  $P \cap A \in \text{Spec}(A)$ , and  $A/(P \cap A) \hookrightarrow B/P$  is an integral extension. Since  $B/P$  is a field,  $A/(P \cap A)$  is a field, so  $P \cap A \in \text{Max}(A)$ .

Conversely, let  $\mathfrak{p} \in \text{Max}(A)$ . It suffices to show  $\mathfrak{p}B \neq B$ : if  $\mathfrak{p}B$  is a proper ideal, there is  $P \in \text{Max}(B)$  with  $\mathfrak{p}B \subseteq P$ , so  $\mathfrak{p} \subseteq P \cap A$ ; since  $1 \notin P \cap A$  and  $\mathfrak{p}$  is maximal,  $P \cap A = \mathfrak{p}$ .

Suppose instead  $\mathfrak{p}B = B$ . Then  $1 = b_1p_1 + \cdots + b_np_n$  for some  $b_i \in B, p_i \in \mathfrak{p}$ . Let  $C = A[b_1, \dots, b_n]$ ; since each  $b_i$  is integral over  $A$ ,  $C$  is a finitely generated  $A$ -module. From  $1 = \sum b_ip_i$  (with  $b_i \in C, p_i \in \mathfrak{p}$ ),  $\mathfrak{p}C = C$ .

By Nakayama's Lemma, there is  $a \in A$  with  $a \equiv 1 \pmod{\mathfrak{p}}$  and  $aC = 0$ ; since  $1 \in C, a = a \cdot 1 = 0$ , so  $0 \equiv 1 \pmod{\mathfrak{p}}$ , i.e.  $1 \in \mathfrak{p}$  – a contradiction.  $\square$

**Theorem 1.65: Lying Over.** Let  $A \hookrightarrow B$  be an integral extension and  $\mathfrak{p} \in \text{Spec}(A)$ .

- (i) There exists  $P \in \text{Spec}(B)$  such that  $P \cap A = \mathfrak{p}$  (we say  $P$  lies over  $\mathfrak{p}$ ).
- (ii) There is no inclusion between distinct prime ideals of  $B$  lying over  $\mathfrak{p}$ : if  $P \subseteq Q$  and  $P \cap A = \mathfrak{p} = Q \cap A$ , then  $P = Q$ .

*Proof.* Localize at  $S = A \setminus \mathfrak{p}$ : we obtain  $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}} := S^{-1}B$ , an integral extension (by the localization-

integral-extension theorem below), with  $A_{\mathfrak{p}}$  local with maximal ideal  $\mathfrak{p}A_{\mathfrak{p}}$ . By the previous lemma (applied to  $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}}$  and the maximal ideal  $\mathfrak{p}A_{\mathfrak{p}}$ ), there is a maximal ideal  $\mathfrak{P}$  of  $B_{\mathfrak{p}}$  with  $\mathfrak{P} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ .

Let  $P \subseteq B$  be the contraction of  $\mathfrak{P}$  along  $B \rightarrow B_{\mathfrak{p}}$ . Then  $P \cap A = \mathfrak{p}$ , proving (i).

Moreover, the primes of  $B$  lying over  $\mathfrak{p}$  are in bijective correspondence with the maximal ideals of  $B_{\mathfrak{p}}$  lying over  $\mathfrak{p}A_{\mathfrak{p}}$  (via this same contraction/extension correspondence). Since distinct maximal ideals are never comparable by inclusion, neither are the corresponding primes of  $B$  lying over  $\mathfrak{p}$ , proving (ii).  $\square$

**Theorem 1.66.** Let  $A \hookrightarrow B$  be an extension of rings and  $C$  the integral closure of  $A$  in  $B$ . If  $S$  is a multiplicative subset of  $A$ , then  $S^{-1}C$  is the integral closure of  $S^{-1}A$  in  $S^{-1}B$ .

*Proof.* Let  $b/s \in S^{-1}B$  be integral over  $S^{-1}A$ :

$$\left(\frac{b}{s}\right)^n + \frac{a_1}{s_1} \left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0.$$

Let  $t = s_1 \cdots s_n$ . Multiplying through by  $(st)^n$ ,

$$(st)^n \left[ \left(\frac{b}{s}\right)^n + \cdots \right] = 0$$

becomes a monic equation in  $bt$  with coefficients in  $A$ . Hence  $bt$  is integral over  $A$ , i.e.  $bt \in C$ , so

$$\frac{b}{s} = \frac{bt}{st} \in S^{-1}C.$$

$\square$

## 1.8 Lecture 8

**Theorem 1.67: Going Up.** Let  $A \hookrightarrow B$  be an integral extension. Let  $\mathfrak{p}_1 \subseteq \mathfrak{p}_2 \subseteq \cdots \subseteq \mathfrak{p}_n$  be a chain in  $\text{Spec } A$ , and let  $Q_1 \subseteq \cdots \subseteq Q_m$  ( $m < n$ ) be a chain in  $\text{Spec } B$  with  $Q_i \cap A = \mathfrak{p}_i$  for all  $i \in [m]$ . Then the chain can be extended to a chain

$$Q_1 \subseteq \cdots \subseteq Q_m \subseteq Q_{m+1} \subseteq \cdots \subseteq Q_n$$

in  $\text{Spec } B$  with  $Q_i \cap A = \mathfrak{p}_i$  for all  $i \in [n]$ .

*Proof.* By induction on  $n - m$ , it suffices to treat the case  $n = 2, m = 1$ : given  $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$  in  $\text{Spec } A$  and  $Q_1 \in \text{Spec } B$  with  $Q_1 \cap A = \mathfrak{p}_1$ , we must find  $Q_2 \in \text{Spec } B$  with  $Q_1 \subseteq Q_2$  and  $Q_2 \cap A = \mathfrak{p}_2$ .

Let  $\bar{A} = A/\mathfrak{p}_1$  and  $\bar{B} = B/Q_1$ . Since  $Q_1 \cap A = \mathfrak{p}_1$ , the induced map  $\bar{A} \rightarrow \bar{B}$  is injective, and by Proposition 1.57(1),  $\bar{A} \hookrightarrow \bar{B}$  is an integral extension.

Now  $\mathfrak{p}_2/\mathfrak{p}_1 \in \text{Spec } \bar{A}$ . By Lying Over (Theorem 1.65), there is  $\bar{Q} \in \text{Spec } \bar{B}$  with  $\bar{Q} \cap \bar{A} = \mathfrak{p}_2/\mathfrak{p}_1$ . By

the correspondence theorem,  $\bar{Q} = Q_2/Q_1$  for a unique  $Q_2 \in \text{Spec } B$  with  $Q_1 \subseteq Q_2$ .

Finally, for  $x \in A$ ,

$$x \in Q_2 \iff \bar{x} \in \bar{Q} \cap \bar{A} = \mathfrak{p}_2/\mathfrak{p}_1 \iff x \in \mathfrak{p}_2,$$

where the last step uses  $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ . Hence  $Q_2 \cap A = \mathfrak{p}_2$ .  $\square$

## 1.9 Lecture 9

**Proposition 1.68.** Let  $\varphi : A \rightarrow B$  be a ring morphism and  $\mathfrak{p} \in \text{Spec } A$ . Then  $\mathfrak{p}$  is the contraction of some  $Q \in \text{Spec } B$  if and only if  $\mathfrak{p}^{ec} = \mathfrak{p}$ .

*Proof.* In general,  $\mathfrak{p} \subseteq \mathfrak{p}^{ec}$ .

( $\Rightarrow$ ) If  $\mathfrak{p} = Q^c$  for some  $Q \in \text{Spec } B$ , then

$$\mathfrak{p}^e = Q^{ce} \subseteq Q \implies \mathfrak{p}^{ec} \subseteq Q^c = \mathfrak{p}.$$

( $\Leftarrow$ ) Assume  $\mathfrak{p}^{ec} = \mathfrak{p}$ . Let  $S = \varphi(A \setminus \mathfrak{p})$ , a multiplicative subset of  $B$ .

**Note:**  $\mathfrak{p}^e \cap S = \emptyset$ . Indeed, if  $x \in \mathfrak{p}^e \cap S$ , then  $\varphi^{-1}(x) \in A \setminus \mathfrak{p}$ ; but  $x \in \mathfrak{p}^e$  gives  $\varphi^{-1}(x) \in \mathfrak{p}^{ec} = \mathfrak{p}$  – a contradiction.

So  $\mathfrak{p}^e S^{-1}B$  is a proper ideal of  $S^{-1}B$ , hence contained in some  $\mathfrak{m} \in \text{Max}(S^{-1}B)$ . Let  $Q$  be the contraction of  $\mathfrak{m}$  in  $B$ ; then  $Q \in \text{Spec } B$  and  $\mathfrak{p}^e \subseteq Q$ . Moreover  $Q \cap S = \emptyset$ , which gives  $Q^c \subseteq \mathfrak{p}$ .

Now  $\mathfrak{p}^{ec} \subseteq Q^c \subseteq \mathfrak{p}$ , and since  $\mathfrak{p}^{ec} = \mathfrak{p}$ , we get  $\mathfrak{p} = Q^c$ .  $\square$

**Proposition 1.69.** Let  $A \hookrightarrow B$  be rings and  $C$  the integral closure of  $A$  in  $B$ . Let  $I \subseteq A$  be an ideal. Then the integral closure of  $I$  in  $B$  is  $\sqrt{IC} \subseteq C$ .

*Proof.* Suppose  $x$  is integral over  $I$  in  $B$ :

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0, \quad a_i \in I.$$

Then  $x \in C$  (a fortiori integral over  $A$ ), and

$$x^n = -(a_1 x^{n-1} + \cdots + a_n) \in IC,$$

so  $x \in \sqrt{IC}$ .

Conversely, let  $x \in \sqrt{IC}$ , so  $x^n \in IC$  for some  $n > 0$ :

$$x^n = a_1 x_1 + \cdots + a_m x_m, \quad a_i \in I, x_i \in C.$$

Each  $x_i \in C$  is integral over  $A$ , so  $M := A[x_1, \dots, x_m]$  is a finitely generated  $A$ -module, and  $x^n M \subseteq IM$ .

Consider the endomorphism  $\mu_{x^n} : M \rightarrow M$  given by multiplication by  $x^n$ ; since  $x^n M \subseteq IM$ , the

determinant-trick argument underlying Nakayama's Lemma (Lecture 6) yields a monic relation

$$(x^n)^k + c_{k-1}(x^n)^{k-1} + \cdots + c_0 = 0, \quad c_i \in I^i,$$

as endomorphisms of  $M$ . Evaluating at  $1 \in M$  (since  $1 \in A \subseteq M$ ) gives this same relation as an identity in  $B$ , exhibiting  $x$  as integral over  $I$ .  $\square$

**Proposition 1.70.** Let  $A \hookrightarrow B$  be integral domains,  $I \subseteq A$  an ideal, and suppose  $A$  is integrally closed with  $Q(A) = K$ . Let  $x \in B$  be integral over  $I$ . Then  $x$  is algebraic over  $K$ , and if

$$\min(x, K) = T^n + a_1 T^{n-1} + \cdots + a_n,$$

then  $a_i \in \sqrt{I}$  for all  $i$ .

*Proof.* Clearly  $x$  is algebraic over  $K$  (being integral over  $A \subseteq K$ ). Let  $L$  be an algebraic extension of  $K$  containing the conjugates  $x_1, \dots, x_n$  of  $x$ .

Each  $x_i$  is integral over  $I$  (as in the proof of the minimal-polynomial theorem from Lecture 7). Using the relations between roots and coefficients,  $a_1, \dots, a_n$  are all integral over  $I$ .

Applying the previous proposition with  $B$  replaced by  $K$  (in which the integral closure of  $A$  is  $A$  itself, since  $A$  is integrally closed), we get  $a_i \in \sqrt{I}$  for all  $i$ .  $\square$

**Theorem 1.71: Going Down.** Assume:

- (a)  $A \hookrightarrow B$  is an integral extension;
- (b)  $A, B$  are integral domains;
- (c)  $A$  is integrally closed.

Let  $\mathfrak{p}_1 \supseteq \mathfrak{p}_2 \supseteq \cdots \supseteq \mathfrak{p}_n$  be a chain in  $\text{Spec } A$ , and let  $Q_1 \supseteq \cdots \supseteq Q_m$  ( $m \leq n$ ) be a chain in  $\text{Spec } B$  with  $Q_i \cap A = \mathfrak{p}_i$  for all  $i \in [m]$ . Then the chain can be extended:

$$Q_1 \supseteq \cdots \supseteq Q_m \supseteq Q_{m+1} \supseteq \cdots \supseteq Q_n$$

with  $Q_i \in \text{Spec } B$  and  $Q_i \cap A = \mathfrak{p}_i$  for all  $i \in [n]$ .

*Proof.* It suffices to treat  $m = 1, n = 2$ : we need  $Q_2 \in \text{Spec } B$  with  $Q_1 \supseteq Q_2$  and  $Q_2 \cap A = \mathfrak{p}_2$ .

The prime ideals of  $B_{Q_1}$  are in bijective correspondence with the primes of  $B$  contained in  $Q_1$ , via

$$A \hookrightarrow B \hookrightarrow B_{Q_1}.$$

It suffices to show  $\mathfrak{p}_2 B_{Q_1} \cap A = \mathfrak{p}_2$ . The containment  $\mathfrak{p}_2 \subseteq \mathfrak{p}_2 B_{Q_1} \cap A$  is clear; we show the reverse.

Let  $x \in \mathfrak{p}_2 B_{Q_1} \cap A$ , so  $x = y/s$  with  $y \in \mathfrak{p}_2 B$ ,  $s \in B \setminus Q_1$ . Write

$$y = r_1 b_1 + \cdots + r_k b_k, \quad r_i \in \mathfrak{p}_2, b_i \in B.$$

Since  $B$  is integral over  $A$ , each  $b_i$  is integral over  $A$ , so  $y$  is integral over  $\mathfrak{p}_2$  (by the previous propositions).

By the proposition on minimal polynomials (applied with  $I = \mathfrak{p}_2$ ),  $\min(y, K)$  (where  $K = \mathbb{Q}(A)$ ) has all coefficients in  $\sqrt{\mathfrak{p}_2} = \mathfrak{p}_2$ :

$$y^r + u_1 y^{r-1} + \cdots + u_r = 0, \quad u_i \in \mathfrak{p}_2.$$

Since  $x = y/s \in A$ , we have  $s = yx^{-1}$  with  $x^{-1} \in K$  (the case  $x = 0$  being trivial). Substituting  $y = sx$  into the relation above and dividing by  $x^r$ ,

$$s^r + \frac{u_1}{x} s^{r-1} + \cdots + \frac{u_r}{x^r} = 0,$$

a minimal equation for  $s$  over  $K$ . Since  $s \in B$  is integral over  $A$ , applying the proposition again (with  $I = A$ ) gives  $u_i/x^i \in \sqrt{A} = A$  for all  $i$ .

Let  $v_i := u_i/x^i \in A$ , so  $v_i x^i = u_i \in \mathfrak{p}_2$ . FTSOC, suppose  $x \notin \mathfrak{p}_2$ . Since  $\mathfrak{p}_2$  is prime,  $x^i \notin \mathfrak{p}_2$ , so  $v_i \in \mathfrak{p}_2$  for all  $i$ . Then

$$s^r = -\left(\frac{u_1}{x} s^{r-1} + \cdots + \frac{u_r}{x^r}\right) \in \mathfrak{p}_2 B \subseteq \mathfrak{p}_1 B \subseteq Q_1,$$

so  $s^r \in Q_1$ ; since  $Q_1$  is prime,  $s \in Q_1$  – contradicting  $s \in B \setminus Q_1$ .

Hence  $x \in \mathfrak{p}_2$ , completing the proof.  $\square$

## Tensor Products of Modules

**Definition 1.72.** Let  $R$  be a commutative ring and  $M, N, P$  be  $R$ -modules. A *tensor product* of  $M, N$  is a pair  $(T, \theta)$ , where  $T$  is an  $R$ -module and  $\theta : M \times N \rightarrow T$  is bilinear, such that for every bilinear map  $f : M \times N \rightarrow P$ , there is a unique linear map  $h : T \rightarrow P$  with

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \theta \downarrow & \nearrow h & \\ T & & \end{array}$$

i.e.  $h \circ \theta = f$ .

**Exercise 1.73.** The tensor product, if it exists, is unique up to isomorphism.

**Remark 1.74.**  $\text{Im}(\theta)$  generates  $T$ .

*Proof.* Let  $T'$  be the submodule of  $T$  generated by  $\text{Im}(\theta)$ , with inclusion  $i : T' \hookrightarrow T$ . Since  $\text{Im}(\theta) \subseteq T'$ ,

$\theta$  corestricts to a bilinear map  $\theta' : M \times N \rightarrow T'$ . By the universal property of  $(T, \theta)$  applied to the bilinear map  $\theta'$  (viewed as landing in  $T$  via  $i \circ \theta' = \theta$ ), there is a unique  $h : T \rightarrow T'$  with  $h \circ \theta = \theta'$ ; composing with  $i$ ,  $(i \circ h) \circ \theta = i \circ \theta' = \theta$ .

But  $\text{id}_T$  also satisfies  $\text{id}_T \circ \theta = \theta$ ; by the uniqueness clause of the universal property,  $i \circ h = \text{id}_T$ . Hence  $i$  is surjective, and since  $i$  is also injective (being an inclusion),  $T' = T$ .  $\square$

**Construction 1.75.** Let  $(F, \iota)$  be the free  $R$ -module on the set  $M \times N$ . Let  $H$  be the submodule of  $F$  generated by all elements of the form

$$(a) \quad \iota(m_1 + m_2, n) - \iota(m_1, n) - \iota(m_2, n),$$

$$(b) \quad \iota(m, n_1 + n_2) - \iota(m, n_1) - \iota(m, n_2),$$

$$(c) \quad \iota(rm, n) - r \iota(m, n),$$

for  $m, m_1, m_2 \in M, n, n_1, n_2 \in N, r \in R$ . Write  $M \otimes_R N := F/H$ , and let  $\pi : F \rightarrow F/H$  be the natural projection.

$\pi \circ \iota$  is bilinear. For instance,

$$(\pi \circ \iota)(m_1 + m_2, n) = \iota(m_1 + m_2, n) + H = (\iota(m_1, n) + \iota(m_2, n)) + H = (\pi \circ \iota)(m_1, n) + (\pi \circ \iota)(m_2, n),$$

using relation (a); the other bilinearity identities follow similarly from (b) and (c).  $\square$

Write  $\otimes_R := \pi \circ \iota : M \times N \rightarrow M \otimes_R N$ , so that  $\otimes_R(m, n) =: m \otimes n$  are the *simple tensors*.

$(M \otimes_R N, \otimes_R)$  satisfies the universal property. Let  $P$  be an  $R$ -module and  $g : M \times N \rightarrow P$  bilinear. Since  $F$  is free on  $M \times N$ , there is a unique morphism  $j : F \rightarrow P$  such that  $j \circ \iota = g$ .

$H \subseteq \ker j$ . For instance,

$$j(\iota(m_1 + m_2, n) - \iota(m_1, n) - \iota(m_2, n)) = g(m_1 + m_2, n) - g(m_1, n) - g(m_2, n) = 0,$$

using bilinearity of  $g$ ; similarly for the other generators of  $H$ .

Hence  $j$  induces a morphism  $\bar{k} : M \otimes_R N \rightarrow P$  with  $\bar{k} \circ \pi = j$ , i.e.

$$\bar{k} \circ \otimes_R = \bar{k} \circ (\pi \circ \iota) = (\bar{k} \circ \pi) \circ \iota = j \circ \iota = g.$$

**Uniqueness.** Let  $k : M \otimes_R N \rightarrow P$  be any morphism with  $k \circ \otimes_R = g$ , i.e.  $k \circ \pi \circ \iota = g = \bar{k} \circ \pi \circ \iota$ . By the uniqueness property of the free module  $F$ ,  $k \circ \pi = \bar{k} \circ \pi$ ; since  $\pi$  is onto,  $k = \bar{k}$ .  $\square$

## 1.10 Lecture 10

**Construction 1.76.** Define  $R \times (R \otimes M) \rightarrow R \otimes M$  by

$$\left( s, \sum_{i=1}^k r_i \otimes m_i \right) \mapsto \sum_{i=1}^k (sr_i) \otimes m_i.$$

One checks this makes  $R \otimes M$  an  $R$ -module.

**Theorem 1.77.**  $R \otimes_R M \cong M$  as  $R$ -modules.

*Proof.* The map  $R \times M \rightarrow M$ ,  $(r, m) \mapsto rm$ , is bilinear, so it induces a unique morphism  $f : R \otimes M \rightarrow M$  with  $f(r \otimes m) = rm$ .

Define  $g : M \rightarrow R \otimes M$  by  $g(m) = 1 \otimes m$ . Then  $f$  is clearly surjective, and  $f, g$  are mutually inverse.  $\square$

From the definition of the tensor product,

$$\{\text{bilinear maps } M \times N \rightarrow P\} \longleftrightarrow \{\text{module morphisms } M \otimes N \rightarrow P\} =: \text{Hom}_R(M \otimes N, P).$$

**Remark 1.78.** If  $M_1, \dots, M_n$  are  $R$ -modules,  $M_1 \otimes M_2 \otimes \dots \otimes M_n$  can be defined in exactly the same way.

**Proposition 1.79.** Let  $M, N, P$  be  $R$ -modules. There are unique isomorphisms such that:

- (a)  $M \otimes N \cong N \otimes M$ ;
- (b)  $(M \otimes N) \otimes P \cong M \otimes (N \otimes P) \cong M \otimes N \otimes P$ , with  $(m \otimes n) \otimes p \mapsto m \otimes (n \otimes p) \mapsto m \otimes n \otimes p$ ;
- (c)  $(M \oplus N) \otimes P \cong (M \otimes P) \oplus (N \otimes P)$ , with  $(m \oplus n) \otimes p \mapsto (m \otimes p) \oplus (n \otimes p)$ ;
- (d)  $R \otimes M \cong M$ , with  $r \otimes m \mapsto rm$ .

*Proof.* (a), (c), (d) are straightforward verifications via the universal property; (b) is similar.  $\square$

**Proposition 1.80.** Let  $x_i \in M$ ,  $y_i \in N$  be such that  $\sum x_i \otimes y_i = 0$  in  $M \otimes N$  (a finite sum). Then there exist finitely generated submodules  $M' \subseteq M$  and  $N' \subseteq N$  such that  $\sum x_i \otimes y_i = 0$  already in  $M' \otimes N'$ .

*Proof.* Recall  $M \otimes N := F/H$ , where  $F$  is free on  $M \times N$ . Since  $\sum x_i \otimes y_i = 0$ , the element  $\sum \iota(x_i, y_i) \in H$  is a finite sum of the generators of  $H$ .

Let  $M'$  be the submodule of  $M$  generated by the  $x_i$ 's together with the first coordinates appearing in these generators of  $H$ , and similarly  $N'$  using the  $y_i$ 's and second coordinates. These are finitely

generated, and the relation exhibiting  $\sum \iota(x_i, y_i) \in H$  only involves elements of  $M'$  and  $N'$ , so  $\sum x_i \otimes y_i = 0$  already in  $M' \otimes N'$ .  $\square$

**Definition 1.81.** An  $R$ -algebra is a set  $S$  such that:

- (1)  $S$  is an  $R$ -module;
- (2)  $S$  is a ring with 1 (associative);
- (3)  $r(s_1 s_2) = (rs_1)s_2 = s_1(rs_2)$  for all  $r \in R, s_1, s_2 \in S$ .

**Example 1.82.** Let  $R, S$  be rings and  $f : R \rightarrow S$  a morphism such that  $f(R) \subseteq Z(S)$  (the center of  $S$ ). Then  $S$  is an  $R$ -algebra, via

$$R \times S \rightarrow S, \quad (r, s) \mapsto f(r) \cdot s.$$

**Example 1.83.** Let  $R, S$  be commutative rings and  $f : R \rightarrow S$  a ring morphism. Then  $S$  is an  $R$ -algebra (a special case of the above).

**Remark 1.84.** Every ring is a  $\mathbb{Z}$ -algebra, via  $f : \mathbb{Z} \rightarrow R, n \mapsto n \cdot 1_R$ .

**Definition 1.85.** Let  $f : R \rightarrow S$  be a ring morphism and  $N$  an  $S$ -module. Define  $R \times N \rightarrow N$  by  $(r, n) \mapsto f(r) \cdot n$ ; this makes  $N$  an  $R$ -module (*restriction of scalars*). In particular,  $S$  is an  $R$ -algebra.

**Definition 1.86.** Let  $M$  be an  $R$ -module and  $S$  an  $R$ -algebra. Define

$$S \times (M \otimes_R S) \rightarrow M \otimes_R S, \quad (s, m \otimes t) \mapsto m \otimes st.$$

Then  $M \otimes_R S$  is an  $S$ -module (*extension of scalars*).

**Remark 1.87.** Let  $M$  be a finitely generated  $R$ -module, generated by  $\{x_1, \dots, x_n\}$ . Then  $M \otimes_R S$  is a finitely generated  $S$ -module, generated by  $\{x_1 \otimes 1, \dots, x_n \otimes 1\}$ .

**Theorem 1.88.**

$$\mathrm{Hom}_R(M \otimes N, P) \cong \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P)).$$

*Proof.* Let  $f : M \times N \rightarrow P$  be bilinear. For each  $x \in M, y \mapsto f(x, y)$  gives a linear map  $N \rightarrow P$ . Hence

$$x \mapsto (y \mapsto f(x, y))$$

is a linear map from  $M$  to  $\text{Hom}_R(N, P)$ .

Conversely, take a morphism  $\varphi : M \rightarrow \text{Hom}_R(N, P)$ . Define  $M \times N \rightarrow P$  by  $(m, n) \mapsto \varphi(m)(n)$ ; this is bilinear.

These two constructions are mutually inverse, giving the desired isomorphism.  $\square$

*Notation. Fact.*  $M' \rightarrow M \rightarrow M'' \rightarrow 0$  is exact if and only if, for every  $R$ -module  $D$ ,

$$0 \rightarrow \text{Hom}(M'', D) \rightarrow \text{Hom}(M, D) \rightarrow \text{Hom}(M', D)$$

is exact.

Let  $f : M \rightarrow M'$  and  $g : N \rightarrow N'$  be morphisms. Define  $M \times N \rightarrow M' \otimes N'$  by  $(m, n) \mapsto f(m) \otimes g(n)$ ; this is bilinear, so it induces a morphism

$$f \otimes g : M \otimes N \rightarrow M' \otimes N', \quad (f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

**Proposition 1.89.** Let  $M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$  be exact, and let  $N$  be any  $R$ -module. Then

$$M' \otimes N \xrightarrow{f \otimes 1} M \otimes N \xrightarrow{g \otimes 1} M'' \otimes N \rightarrow 0$$

is exact.

*Proof.* Call the first sequence  $E$  and the second  $E \otimes N$ . Let  $P$  be an  $R$ -module.

$E$  exact implies

$$0 \rightarrow \text{Hom}(M'', \text{Hom}(N, P)) \rightarrow \text{Hom}(M, \text{Hom}(N, P)) \rightarrow \text{Hom}(M', \text{Hom}(N, P))$$

is exact (by the Fact above, applied with  $D = \text{Hom}(N, P)$ ), which (using the Tensor-Hom adjunction) gives

$$0 \rightarrow \text{Hom}(M'' \otimes N, P) \rightarrow \text{Hom}(M \otimes N, P) \rightarrow \text{Hom}(M' \otimes N, P)$$

exact. By the Fact again,  $M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$  is exact.  $\square$

**Theorem 1.90.** Let  $R$  be a ring,  $S \subseteq R$  multiplicative, and  $M$  an  $R$ -module. Then

$$S^{-1}R \otimes_R M \cong S^{-1}M.$$

*Proof.* Define  $S^{-1}R \times M \rightarrow S^{-1}M$  by  $(a/s, m) \mapsto am/s$ ; this is  $R$ -linear. This induces a morphism

$$f : S^{-1}R \otimes_R M \rightarrow S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s},$$

uniquely determined, and clearly surjective.

**Injectivity.** Take a typical element of  $S^{-1}R \otimes_R M$ ,

$$\frac{a_1}{s_1} \otimes m_1 + \cdots + \frac{a_n}{s_n} \otimes m_n.$$

Let  $s = s_1 \cdots s_n$  and  $t_j = \prod_{i \neq j} s_i$ , so  $a_j/s_j = a_j t_j/s$ . Then

$$\frac{a_j}{s_j} \otimes m_j = \frac{a_j t_j}{s} \otimes m_j = \frac{1}{s} \otimes (a_j t_j m_j),$$

so the element becomes

$$\frac{1}{s} \otimes (a_1 t_1 m_1) + \cdots + \frac{1}{s} \otimes (a_n t_n m_n) = \frac{1}{s} \otimes m, \quad m := a_1 t_1 m_1 + \cdots + a_n t_n m_n.$$

Suppose  $f(1/s \otimes m) = 0$ , i.e.  $m/s = 0$  in  $S^{-1}M$ . Then there is  $t \in S$  with  $tm = 0$ , and

$$\frac{1}{s} \otimes m = \frac{t}{st} \otimes m = \frac{1}{st} \otimes (tm) = \frac{1}{st} \otimes 0 = 0.$$

Hence  $f$  is injective, so  $S^{-1}R \otimes_R M \cong S^{-1}M$ . □

## 1.11 Lecture 11

### Chain Conditions

**Proposition 1.91.** Let  $(\Sigma, \leq)$  be a poset. Then the following are equivalent:

- (1) Every ascending chain  $x_1 \leq x_2 \leq \cdots$  in  $\Sigma$  is stationary.
- (2) Every nonempty subset  $T \subseteq \Sigma$  has a maximal element.

*Proof.* (1)  $\Rightarrow$  (2): If  $\emptyset \neq T \subseteq \Sigma$  had no maximal element, we could pick  $x_1 \in T$  and inductively  $x_{n+1} \in T$  with  $x_n < x_{n+1}$ , giving a strictly ascending chain which is not stationary.

(2)  $\Rightarrow$  (1): Given a chain  $x_1 \leq x_2 \leq \cdots$ , the set  $\{x_n\}_{n \geq 1}$  has a maximal element  $x_k$ ; then  $x_n = x_k$  for all  $n \geq k$ . □

**Definition 1.92.** An  $R$ -module  $M$  is called *noetherian* if every ascending chain of submodules of  $M$  is stationary. A ring  $R$  is *noetherian* if  $R$  is a noetherian  $R$ -module.

**Theorem 1.93.** An  $R$ -module  $M$  is noetherian if and only if every submodule of  $M$  is finitely generated.

*Proof.* ( $\Rightarrow$ ) Let  $M$  be noetherian and  $N$  a submodule of  $M$ . Define

$$\Sigma = \{\text{all finitely generated submodules of } N\}.$$

Since  $\{0\} \in \Sigma$ ,  $\Sigma \neq \emptyset$ . As  $N$  is noetherian (a chain of submodules of  $N$  is a chain in  $M$ ),  $\Sigma$  has a maximal element  $N'$  by Proposition 1.91. It is enough to show  $N' = N$ .

If  $N \neq N'$ , there is  $x \in N$  with  $x \notin N'$ . Then  $N' + xR \supsetneq N'$ , and  $N' + xR$  is also finitely generated, contradicting the maximality of  $N'$ .

( $\Leftarrow$ ) Suppose every submodule of  $M$  is finitely generated, and let  $M_1 \subseteq M_2 \subseteq \cdots$  be an ascending chain of submodules of  $M$ . Then  $N = \bigcup_{n \geq 1} M_n$  is a submodule, so by hypothesis it is finitely generated, say

$$N = x_1R + x_2R + \cdots + x_nR,$$

with each  $x_j \in M_{i_j}$  for some  $i_j$ . Taking  $k = \max_j i_j$ , we have  $x_1, \dots, x_n \in M_k$ , so  $N \subseteq M_k$  and hence  $M_k = N$ . Therefore

$$M_k = M_{k+1} = M_{k+2} = \cdots = N. \quad \square$$

**Theorem 1.94.** Let  $0 \rightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \rightarrow 0$  be an exact sequence of  $R$ -modules. Then  $M$  is noetherian if and only if  $M'$  and  $M''$  are noetherian.

*Proof.* ( $\Rightarrow$ ) Let  $M$  be noetherian, so  $M$  satisfies the ascending chain condition on submodules.

Let  $N'_1 \subseteq N'_2 \subseteq \cdots$  be an ascending chain in  $M'$ . Then  $\alpha(N'_1) \subseteq \alpha(N'_2) \subseteq \cdots$  is an ascending chain in  $M$ , hence stationary: there is  $k$  with  $\alpha(N'_k) = \alpha(N'_{k+i})$  for all  $i \geq 0$ . As  $\alpha$  is injective,  $N'_k = N'_{k+i}$  for all  $i \geq 0$ . So  $M'$  is noetherian.

Next,  $M'' \cong M / \text{Ker } \beta$ . Any ascending chain in  $M''$  is of the form

$$N_1 / \text{Ker } \beta \subseteq N_2 / \text{Ker } \beta \subseteq \cdots,$$

where each  $N_i$  is a submodule of  $M$  containing  $\text{Ker } \beta$ , and  $N_1 \subseteq N_2 \subseteq \cdots$ . The latter chain is stationary in  $M$ , so the chain  $N_1 / \text{Ker } \beta \subseteq N_2 / \text{Ker } \beta \subseteq \cdots$  is also stationary.

( $\Leftarrow$ ) Conversely, let  $M', M''$  be noetherian, and take an ascending chain  $N_1 \subseteq N_2 \subseteq \cdots$  in  $M$ . Then

$$\alpha^{-1}(N_1) \subseteq \alpha^{-1}(N_2) \subseteq \cdots \text{ in } M', \quad \beta(N_1) \subseteq \beta(N_2) \subseteq \cdots \text{ in } M'',$$

and both are stationary. Taking the bigger of the two indices, there is  $n$  such that

$$\alpha^{-1}(N_n) = \alpha^{-1}(N_{n+1}) = \cdots \quad \text{and} \quad \beta(N_n) = \beta(N_{n+1}) = \cdots.$$

**Claim:**  $N_n = N_{n+1} = \cdots$ .

It is enough to check  $N_{n+1} \subseteq N_n$  (and likewise for later indices). Take  $x \in N_{n+1}$ . Then  $\beta(x) \in \beta(N_{n+1}) = \beta(N_n)$ , so there is  $y \in N_n$  with  $\beta(y) = \beta(x)$ . Hence

$$x - y \in \text{Ker } \beta = \text{Im } \alpha,$$

so  $x - y = \alpha(z)$  for some  $z \in M'$ . Since  $x - y \in N_{n+1}$ , we get  $z \in \alpha^{-1}(N_{n+1}) = \alpha^{-1}(N_n)$ , so  $\alpha(z) \in N_n$ , i.e.  $x - y \in N_n$ . Thus  $x \in N_n$ , and  $N_{n+1} \subseteq N_n$ .  $\square$

**Remark 1.95.** If  $M_1, M_2$  are noetherian, then  $M_1 \oplus M_2$  is noetherian (apply the theorem to  $0 \rightarrow M_1 \rightarrow M_1 \oplus M_2 \rightarrow M_2 \rightarrow 0$ ), and inductively  $M_1 \oplus M_2 \oplus \cdots \oplus M_n$  is noetherian.

**Corollary 1.96.** Let  $R$  be a noetherian ring. An  $R$ -module  $M$  is noetherian if and only if  $M$  is finitely generated.

*Proof.* ( $\Rightarrow$ ) Immediate from Theorem 1.93 ( $M$  is a submodule of itself).

( $\Leftarrow$ ) Let  $M$  be finitely generated. Then there is a surjective morphism  $f : R^n \rightarrow M$ . Since  $R^n$  is noetherian (a direct sum of noetherian modules), and  $M \cong R^n / \text{Ker } f$ ,  $M$  is noetherian by Theorem 1.94.  $\square$

**Proposition 1.97.** Let  $R$  be noetherian and  $I \subseteq R$  an ideal. Then  $R/I$  is a noetherian ring.

*Proof.* The sequence  $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$  is exact. Since  $R$  is noetherian,  $I$  and  $R/I$  are noetherian  $R$ -modules. The  $R/I$ -submodules of  $R/I$  are exactly its  $R$ -submodules, so  $R/I$  is a noetherian  $R/I$ -module, i.e. a noetherian ring.  $\square$

**Corollary 1.98.** Let  $A \hookrightarrow B$  be rings. Assume  $A$  is noetherian and  $B$  is a finitely generated  $A$ -module. Then  $B$  is a noetherian ring.

*Proof.*  $B$  is a finitely generated  $A$ -module and  $A$  is noetherian, so  $B$  is a noetherian  $A$ -module. Let  $I \subseteq B$  be an ideal. Note that  $I$  is an  $A$ -submodule of  $B$ , so  $I$  is a finitely generated  $A$ -module, hence also a finitely generated  $B$ -module (the same generators work).  $\square$

**Theorem 1.99: Hilbert Basis Theorem.** If  $R$  is noetherian, then  $R[X]$  is also noetherian.

*Proof.* Let  $I \subseteq R[X]$  be an ideal; we show that  $I$  is finitely generated. Consider

$$\ell(I) := \{\text{leading coefficients of elements of } I\} \cup \{0\}.$$

Then  $\ell(I) \subseteq R$  is an ideal. Since  $R$  is noetherian,  $\ell(I)$  is finitely generated, say  $\ell(I) = \langle a_1, \dots, a_n \rangle$ . Let  $f_i \in I$  be such that the leading coefficient of  $f_i$  is  $a_i$ , and let

$$I' = \langle f_1, \dots, f_n \rangle \subseteq I.$$

Take  $r := \max_i \{\deg f_i\}$ , and let  $M$  be the  $R$ -module generated by  $\{1, X, X^2, \dots, X^{r-1}\}$ .

**Claim:**  $I = (I \cap M) + I'$ .

It is enough to show  $I \subseteq (I \cap M) + I'$ . Take  $f(X) \in I$ ,  $f(X) = aX^m + (\text{lower degree terms})$ . If  $m < r$ , then  $f(X) \in I \cap M$ . Let  $m \geq r$ . Then  $a \in \ell(I) = \langle a_1, \dots, a_n \rangle$ , so

$$a = u_1 a_1 + \cdots + u_n a_n, \quad u_i \in R.$$

Consider

$$f(X) - \sum_i u_i f_i X^{m-r_i}, \quad r_i = \deg f_i.$$

This lies in  $I$  and reduces the degree of  $f$ . Continuing this way, we find  $g \in I'$  such that  $f - g$  is a polynomial of degree  $\leq r - 1$ . Then

$$f - g \in I \cap M \quad \text{and} \quad g \in I',$$

so  $f \in (I \cap M) + I'$ , proving the claim.

Now  $M$  is a finitely generated  $R$ -module and  $R$  is noetherian, so  $M$  is a noetherian  $R$ -module (Corollary 1.96). Thus its submodule  $I \cap M$  is finitely generated, and  $I'$  is finitely generated. Hence  $I = (I \cap M) + I'$  is finitely generated.  $\square$

**Corollary 1.100.** If  $R$  is noetherian, then  $R[X_1, \dots, X_n]$  is noetherian.

**Corollary 1.101.** If  $R$  is noetherian, then  $R[X_1, \dots, X_n]/I$  is noetherian for every ideal  $I$ . (Such rings are called *finitely generated  $R$ -algebras*.)

**Corollary 1.102.** If  $k$  is a field, then  $k[X_1, \dots, X_n]/I$  is noetherian.

**Theorem 1.103.** A ring  $R$  is noetherian if and only if every prime ideal of  $R$  is finitely generated.

*Proof.*  $(\Rightarrow)$  is clear. For  $(\Leftarrow)$ , let

$$\Sigma := \{K \subseteq R \text{ ideal} \mid K \text{ is not finitely generated}\},$$

and assume  $\Sigma \neq \emptyset$ . The union of a chain in  $\Sigma$  is an ideal, and it is not finitely generated (otherwise its finitely many generators would lie in a single member of the chain, making that member finitely generated). So by Zorn's Lemma,  $\Sigma$  has a maximal element, say  $I$ .

Note that  $I$  is not finitely generated (in particular  $I \neq R$ ) and, by hypothesis, not prime. Hence there exist  $x \notin I, y \notin I$  with  $xy \in I$ .

Take  $I + \langle x \rangle \supsetneq I$ . Then  $I + \langle x \rangle \notin \Sigma$ , so  $I + \langle x \rangle$  is finitely generated. Also consider

$$(I : x) = \{a \in R \mid ax \in I\},$$

which is an ideal. Since  $y \in (I : x) \setminus I$ , we have  $I \subsetneq (I : x)$ , so  $(I : x)$  is finitely generated.

Now write  $I + \langle x \rangle = R\alpha_1 + \dots + R\alpha_n + Rx$  with  $\alpha_i \in I$  (each generator of  $I + \langle x \rangle$  has the form  $\alpha + rx$  with  $\alpha \in I$ , and we may replace it by  $\alpha$ ). Define

$$J := \langle \alpha_1, \dots, \alpha_n \rangle, \quad \text{so that } J + \langle x \rangle = I + \langle x \rangle.$$

**Claim:**  $I = J + x(I : x)$ .

Clearly  $J \subseteq I$  and  $x(I : x) \subseteq I$ . Conversely, let  $a \in I$ . Then  $a \in J + \langle x \rangle$ , so  $a = j + rx$  with  $j \in J$ ,  $r \in R$ . Now  $rx = a - j \in I$ , so  $r \in (I : x)$ , and  $a \in J + x(I : x)$ .

Since  $J$  and  $(I : x)$  are finitely generated, so is  $I = J + x(I : x)$ . This contradicts  $I \in \Sigma$ . Hence  $\Sigma = \emptyset$ , and every ideal of  $R$  is finitely generated.  $\square$

## 1.12 Lecture 12

**Proposition 1.104.** Let  $A \subseteq B \subseteq C$  be rings. Assume

- (1)  $A$  is noetherian;
- (2)  $C$  is a finitely generated  $A$ -algebra;
- (3)  $C$  is a finitely generated  $B$ -module.

Then  $B$  is a finitely generated  $A$ -algebra.

*Proof.* Say  $C = A[\alpha_1, \dots, \alpha_m]$  with  $\alpha_i \in C$ , and let  $\beta_1, \dots, \beta_n$  generate  $C$  as a  $B$ -module, so  $C = B\beta_1 + \dots + B\beta_n$ . Then

$$\alpha_i = \sum_j b_{ij} \beta_j, \quad b_{ij} \in B, \quad (1.1)$$

$$\beta_i \beta_j = \sum_k b_{ijk} \beta_k, \quad b_{ijk} \in B. \quad (1.2)$$

Let  $B_0$  be the  $A$ -algebra generated by the  $b_{ij}$  and the  $b_{ijk}$ . Then  $B_0$  is a finitely generated  $A$ -algebra and  $A$  is noetherian, so  $B_0$  is noetherian (Hilbert basis theorem), and

$$A \subseteq B_0 \subseteq B.$$

An element of  $C$  is a polynomial expression in  $\alpha_1, \dots, \alpha_m$  with coefficients in  $A$ . Substituting (1.1) and then repeatedly applying (1.2) to reduce products of the  $\beta$ 's, we get that

$$C = B_0 + B_0\beta_1 + \dots + B_0\beta_n,$$

so  $C$  is a finitely generated  $B_0$ -module. Since  $B_0$  is noetherian,  $C$  is a noetherian  $B_0$ -module. As  $B$  is a  $B_0$ -submodule of  $C$ , it follows that  $B$  is a finitely generated  $B_0$ -module.

Combining:  $B_0$  is a finitely generated  $A$ -algebra and  $B$  is a finitely generated  $B_0$ -module, so  $B$  is a finitely generated  $A$ -algebra (generated by the algebra generators of  $B_0$  together with the module generators of  $B$  over  $B_0$ ).  $\square$

**Theorem 1.105: Artin–Tate: weak form of Hilbert’s Nullstellensatz.** Let  $k$  be a field and  $E$  a finitely generated  $k$ -algebra. If  $E$  is a field, then  $E$  is a finite algebraic extension of  $k$ .

**Corollary 1.106.** Let  $k$  be a field,  $A$  a finitely generated  $k$ -algebra, and  $\mathfrak{m} \in \text{Max}(A)$ . Then  $A/\mathfrak{m}$  is a finite algebraic extension of  $k$ . In particular, if  $k$  is algebraically closed, then  $A/\mathfrak{m} \cong k$ .

*Proof.*  $A/\mathfrak{m}$  is a field and a finitely generated  $k$ -algebra, so the theorem applies. If  $k$  is algebraically closed, its only finite algebraic extension is  $k$  itself.  $\square$

*Proof of Theorem 1.105.* Let  $E = k[\alpha_1, \dots, \alpha_m]$ . Suppose  $E$  is not algebraic over  $k$ . Then we can rename the  $\alpha_i$  so that  $\alpha_1, \dots, \alpha_r$  ( $r \geq 1$ ) are algebraically independent over  $k$ , i.e. there is no nonzero polynomial  $f \in k[X_1, \dots, X_r]$  with  $f(\alpha_1, \dots, \alpha_r) = 0$ , and each of  $\alpha_{r+1}, \dots, \alpha_m$  is algebraic over

$$F = k(\alpha_1, \dots, \alpha_r).$$

Then  $E = F[\alpha_{r+1}, \dots, \alpha_m]$  is a finite extension of  $F$ , so  $E$  is a finitely generated  $F$ -module. Apply Proposition 1.104 to

$$k \subseteq F \subseteq E$$

( $k$  is noetherian,  $E$  is a finitely generated  $k$ -algebra,  $E$  is a finitely generated  $F$ -module):  $F$  is a finitely generated  $k$ -algebra, say  $F = k[\beta_1, \dots, \beta_s]$ , where

$$\beta_i = \frac{f_i}{g_i}, \quad f_i, g_i \in k[\alpha_1, \dots, \alpha_r], \quad g_i \neq 0.$$

Now  $k[\alpha_1, \dots, \alpha_r] \cong k[X_1, \dots, X_r]$  is a UFD with infinitely many (non-associate) irreducibles, so there is an irreducible  $h$  that is coprime to each  $g_i$ : take  $h$  to be an irreducible factor of  $g_1 \cdots g_s + 1$ . Then  $h \in k[\alpha_1, \dots, \alpha_r]$  and  $h^{-1} \in k(\alpha_1, \dots, \alpha_r) = F$ .

But  $h^{-1}$  cannot be written as a polynomial in the  $\beta_i$ ’s: a polynomial in the  $f_i/g_i$  has the form  $p/(g_1 \cdots g_s)^N$  with  $p \in k[\alpha_1, \dots, \alpha_r]$ , and  $h^{-1} = p/(g_1 \cdots g_s)^N$  would give  $(g_1 \cdots g_s)^N = hp$ , so the irreducible (hence prime)  $h$  would divide some  $g_i$ . Contradiction. Hence  $E$  is algebraic over  $k$ , and being finitely generated by algebraic elements,  $E$  is a finite extension of  $k$ .  $\square$

**Remark 1.107.** The choice of  $h$  needs  $g_1 \cdots g_s + 1$  to be nonconstant. If every  $g_i$  were a constant, then  $F = k[\beta_1, \dots, \beta_s] \subseteq k[\alpha_1, \dots, \alpha_r]$ , a polynomial ring in  $r \geq 1$  variables, which is not a field (e.g.  $\alpha_1$  is not invertible). So we may assume some  $g_i$  is nonconstant, and then  $g_1 \cdots g_s + 1$  is nonconstant.



## CHAPTER 2

# PROBLEMS

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### 2.1 Assignment 1

*Convention.* By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let  $F$  be a field. Prove that  $F[X]$  has infinitely many irreducible polynomials.
2. Let  $R$  be a ring. Show that  $R[X]$  is never a local ring.
3. Let  $R$  be a ring in which every element is either nilpotent or a unit. Prove that  $R$  is a local ring. How many prime ideals can  $R$  have?
4. Let  $R$  be a ring. Prove that  $a \in J(R)$  if and only if  $1 + ab$  is a unit for all  $b \in R$ . (In particular,  $1 + a$  is a unit.)
5. Let  $R$  be a ring and  $I$  be a finitely generated ideal such that  $I^2 = I$ . Show that there exists  $e \in I$  such that  $I = \langle e \rangle$  where  $e^2 = e$  (such an element is called idempotent).
6. Let  $R$  be a ring and  $I$  be a finitely generated ideal such that  $I = \langle a_1, \dots, a_n \rangle + I^2$ . Show that there exists  $e \in I^2$  such that  $I = \langle a_1, \dots, a_n, e \rangle$  where  $e(1 - e) \in \langle a_1, \dots, a_n \rangle$ . What happens if  $R$  is a local ring?
7. Is  $R[X] \hookrightarrow R[X, X^{-1}]$  an integral extension?
8. Let  $A \hookrightarrow B$  be an integral extension where  $A, B$  are integral domains. Let  $C$  be an integral domain and  $\varphi : B \rightarrow C$  be a morphism such that  $\varphi|_A$  is injective. Prove that  $\varphi$  is injective.
9. Determine the integral closure of  $\mathbb{Z}[\sqrt{3}]$  in  $\mathbb{Q}(\sqrt{3})$ .
10. Determine the integral closure of  $\mathbb{Z}[\sqrt{5}]$  in  $\mathbb{Q}(\sqrt{5})$ .

### 2.2 Assignment 2

*Convention.* By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let  $p \in \mathbb{Z}$  be a prime and write  $\mathfrak{p} = \langle p \rangle$ . Describe the elements of the local ring  $\mathbb{Z}_{\mathfrak{p}}$ . What is the field of fractions of  $\mathbb{Z}_{\mathfrak{p}}$ ?

2. If  $N, P$  are submodules of an  $R$ -module  $M$ , prove that:

- (a)  $S^{-1}(N + P) = S^{-1}N + S^{-1}P$ ;
- (b)  $S^{-1}(N \cap P) = S^{-1}N \cap S^{-1}P$ ;
- (c)  $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$  as  $S^{-1}R$ -modules.

3. Let  $R$  be a ring and  $S \subset R$  be a multiplicative subset. We have the canonical ring morphism  $f : R \rightarrow S^{-1}R$  given by  $f(r) = \frac{r}{1}$ . For an ideal  $I$  of  $R$ ,  $I^e$  denotes the ideal in  $S^{-1}R$  generated by  $f(I)$  (this is called *extension* of  $I$ ). If  $J \subseteq S^{-1}R$  is an ideal, then  $f^{-1}(J)$  is an ideal of  $R$  which will be denoted as  $J^c$  (called the *contraction* of  $J$ ).

- (a) Prove that any ideal of  $S^{-1}R$  is an extended ideal.
- (b) Prove that an ideal  $I$  of  $R$  is contraction of an ideal of  $S^{-1}R$  if and only if no element of  $S$  is a zero-divisor in  $R/I$ .
- (c)  $\text{Spec}(S^{-1}R)$  is in bijective correspondence with  $X = \{\mathfrak{p} \in \text{Spec}(R) \mid \mathfrak{p} \cap S = \emptyset\}$ .
- (d) For ideals  $I, J$  of  $R$ , show that:  $S^{-1}(I + J) = S^{-1}I + S^{-1}J$ ,  $S^{-1}(I \cap J) = S^{-1}I \cap S^{-1}J$ ,  $S^{-1}(IJ) = S^{-1}I S^{-1}J$ ,  $S^{-1}(\sqrt{I}) = \sqrt{S^{-1}I}$ .
- (e) If  $N$  is the nilradical of  $R$  then prove that the nilradical of  $S^{-1}R$  is  $S^{-1}N$ .

4. Let  $\mathfrak{p} \in \text{Spec}(R)$ . Then  $R_{\mathfrak{p}}$  is a local ring with unique maximal ideal  $\mathfrak{p}R_{\mathfrak{p}}$  (we proved this). Therefore,  $\frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}}$  is a field, called the *residue field* at  $\mathfrak{p}$ . On the other hand,  $R/\mathfrak{p}$  is an integral domain. Prove that its field of fractions is the same as  $\frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}}$ .

5. Let  $M$  be a finitely generated  $R$ -module and  $S$  be a multiplicative subset of  $R$ . Prove that  $S^{-1}(\text{Ann}(M)) = \text{Ann}(S^{-1}M)$ .

6. Let  $M$  be a finitely generated  $R$ -module and  $S$  be a multiplicative subset of  $R$ . Prove that  $S^{-1}M = 0$  if and only if there is  $s \in S$  such that  $sM = 0$ .

7. Let  $J$  be an ideal of  $R$ . Show that  $S := 1 + J := \{1 + a \mid a \in J\}$  is a multiplicative set. Prove that  $S^{-1}J$  is contained in the Jacobson radical of  $S^{-1}R$ .

8. Let  $R$  be a ring and  $S, T$  be multiplicative subsets. Let  $U$  be the image of  $T$  in  $S^{-1}R$ . Define  $ST := \{st \mid s \in S, t \in T\}$ . Check that  $ST$  is also a multiplicative subset. Prove that the rings  $(ST)^{-1}R$  and  $U^{-1}(S^{-1}R)$  are isomorphic.

9. Let  $R$  be a ring such that  $R_{\mathfrak{p}}$  has no nonzero nilpotent for every  $\mathfrak{p} \in \text{Spec}(R)$ . Prove that  $R$  has no nonzero nilpotent. If  $R$  is a ring such that  $R_{\mathfrak{p}}$  is an integral domain for every  $\mathfrak{p} \in \text{Spec}(R)$ , then does it imply that  $R$  is an integral domain?

10. Let  $R$  be a (nonzero) ring and  $X$  be the set of all multiplicative subsets  $S$  of  $R$  such that  $0 \notin S$ . Show that  $X$  has a maximal element. Prove further that  $T \in X$  is a maximal element of  $X$  if and only if  $R \setminus T$  is a minimal prime ideal of  $R$ .

## 2.3 Assignment 3

*Convention.* By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let  $R$  be an integral domain. Prove that the following are equivalent:
  - (a)  $R$  is integrally closed.
  - (b)  $R_{\mathfrak{p}}$  is integrally closed for all  $\mathfrak{p} \in \text{Spec}(R)$ .
  - (c)  $R_{\mathfrak{m}}$  is integrally closed for all  $\mathfrak{m} \in \text{Max}(R)$ .
2. Let  $R \hookrightarrow S$  be an integral extension.
  - (a) Let  $a \in R$  be such that  $a$  is a unit in  $S$ . Then show that  $a$  is a unit in  $R$ .
  - (b) Prove that  $\text{Jac}(R) = \text{Jac}(S) \cap R$ .
3. Let  $R$  be a ring and  $G$  be a finite subgroup of the group of automorphisms of  $R$  (automorphism means ring isomorphism from  $R$  to  $R$ , and they form a group with respect to composition). Let

$$R^G := \{r \in R \mid \sigma(r) = r \forall \sigma \in G\}.$$

- (a) Check that  $R^G$  is a subring of  $R$  and  $R$  is integral over  $R^G$ .
  - (b) Let  $P \in \text{Spec}(R^G)$  and  $X := \{Q \in \text{Spec}(R) \mid Q \cap R^G = P\}$ . If  $Q_1, Q_2 \in X$ , prove that there is some  $\tau \in G$  such that  $Q_1 = \tau(Q_2)$ . Conclude that  $X$  is finite.
4. Let  $R \hookrightarrow S$  be an integral extension and  $I$  be an ideal of  $R$ . Prove that

$$\sqrt{IS} = \{s \in S \mid s^n + a_1 s^{n-1} + \cdots + a_n = 0 \text{ for some } n \in \mathbb{N} \text{ and } a_i \in I \forall i\}.$$

5. Let  $R \hookrightarrow S$  be an integral extension and let  $\mathfrak{n} \in \text{Max}(B)$ . Let  $\mathfrak{m} = \mathfrak{n} \cap A$  (note that  $\mathfrak{m} \in \text{Max}(A)$ ). Is  $B_{\mathfrak{n}}$  integral over  $A_{\mathfrak{m}}$ ?
6. Let  $R \hookrightarrow S$  be an extension of rings such that  $B \setminus A$  is closed under multiplication. Prove that  $A$  is integrally closed in  $B$ .
7. Let  $R$  be a ring and  $f \in R[X]$  be a nonconstant monic. Show that there is a ring  $S$  that contains  $R$  as a subring such that  $f$  is a product of monic linear factors in  $S[X]$ .
8. Let  $R \hookrightarrow S$  be an extension of rings. Let  $f, g$  be two monic polynomials in  $S[X]$ . Let the coefficients of  $fg$  are all integral over  $R$ . Prove that the all coefficients of  $f$  and all the coefficients of  $g$  are integral over  $R$ .
9. Let  $R \hookrightarrow S$  be an extension of rings. Prove that  $f \in S[X]$  is integral over  $R[X]$  if and only if all the coefficients of  $f$  are integral over  $R$ . Conclude that if  $C$  is the integral closure of  $R$  in  $S$ , then  $C[X]$  is the integral closure of  $R[X]$  in  $S[X]$ .

10. Let  $A$  be an integrally closed domain with field of fractions  $K$ . Let  $L$  be a finite Galois extension of  $K$  with Galois group  $G$ . Let  $B$  be the integral closure of  $A$  in  $L$ . Show that  $\sigma(B) = B$  for all  $\sigma \in G$  and that  $A = B^G = \{b \in B \mid \sigma(b) = b \forall \sigma \in G\}$ .

## 2.4 Assignment 4

*Convention.* By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Show that  $2 \otimes \bar{1}$  is 0 in  $\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{2\mathbb{Z}}$  but is not 0 in  $2\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{2\mathbb{Z}}$ .
2. Using the above exercise, or otherwise, prove that tensor product does not preserve exactness.
3. Let  $a, b \in \mathbb{Z}$  be coprime. Prove that  $\frac{\mathbb{Z}}{a\mathbb{Z}} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{b\mathbb{Z}} = 0$ . More generally, if the gcd of  $a, b$  is  $d$ , prove that
 
$$\frac{\mathbb{Z}}{a\mathbb{Z}} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{b\mathbb{Z}} \cong \frac{\mathbb{Z}}{d\mathbb{Z}}.$$
4. Let  $R$  be a ring and  $I, J$  be ideals of  $R$ . Prove that  $(R/I) \otimes_R (R/J) \cong R/(I+J)$ .
5. Let  $M$  be an  $R$ -module and  $I \subseteq R$  be an ideal. Prove that  $(R/I) \otimes_R M \cong M/IM$ .
6. Prove that  $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$ . Prove also that  $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$ .
7. Let  $M$  be an  $R$ -module and  $I \subseteq R$  be an ideal. Prove that there is an  $R$ -linear map  $g : I \otimes_R M \rightarrow IM$  and that  $g$  is surjective. Give an example to show that  $g$  is not injective in general.
8. Let  $R \hookrightarrow S$  be rings. Prove that  $S \otimes_R R[X]$  is isomorphic to  $S[X]$  as  $R$ -modules.
9. Let  $R$  be an integral domain with quotient field  $K$  and let  $V$  be a  $K$ -vector space. Prove that  $K \otimes_R V \cong V$  as  $R$ -modules.
10. Let  $M, N, P$  be  $R$ -modules. Prove that  $M \otimes (N \otimes P) \cong M \otimes N \otimes P \cong (M \otimes N) \otimes P$ .

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